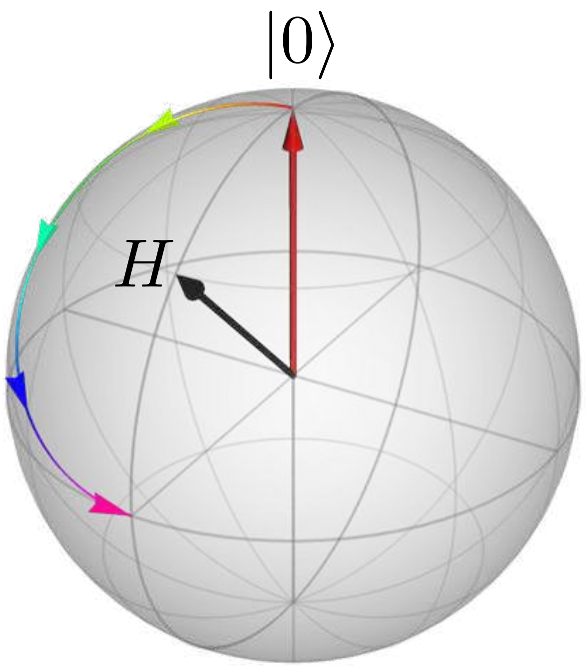
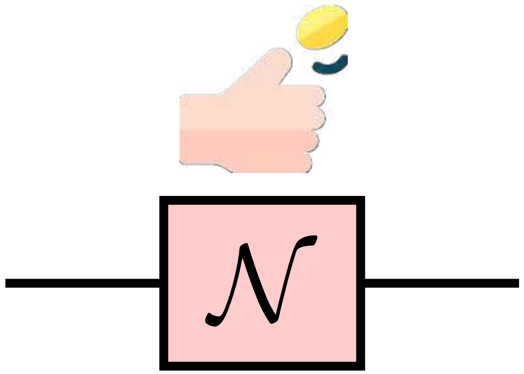
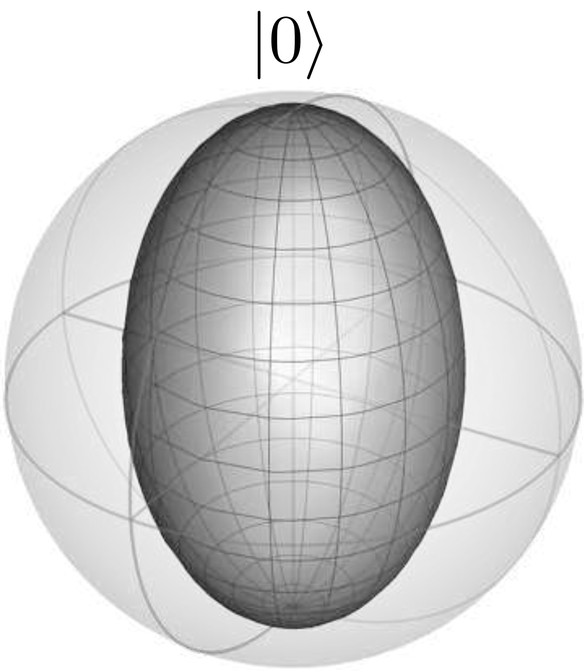
Introduction to Quantum Noise

Qiskit Global Summer School:
Quantum Machine Learning







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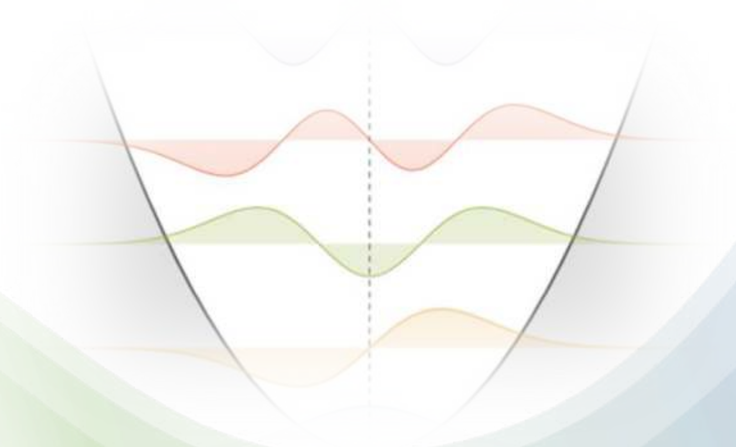


"Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory."

Asher Peres

Hello World! to a real quantum experiment

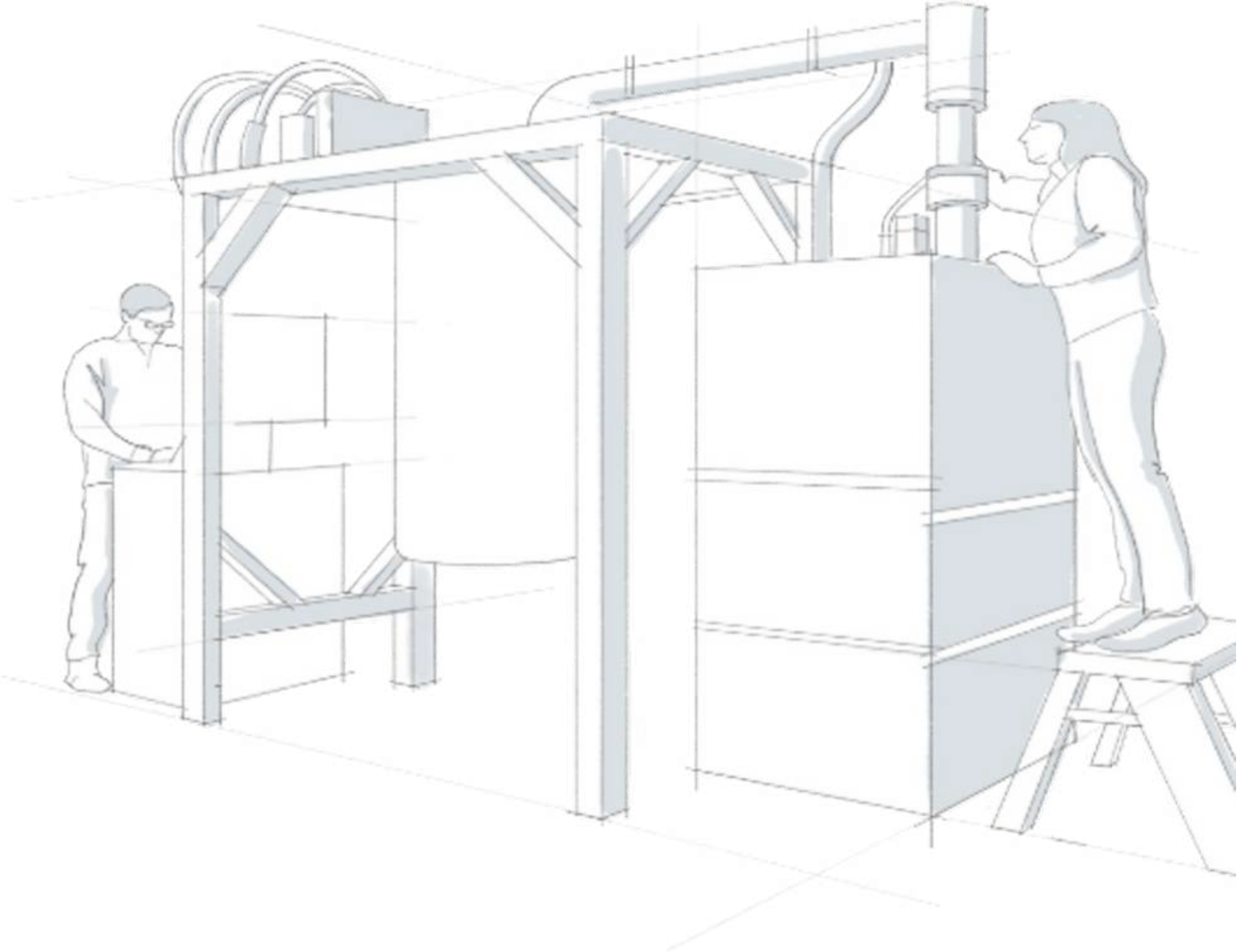


image: qiskit

Hello World! to a real quantum experiment

Qubit computational basis states

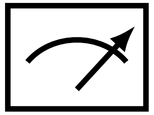
$$|0\rangle\,,|1\rangle$$

Qubit gate

$$\boxed{X}$$

Qubit observable

Z



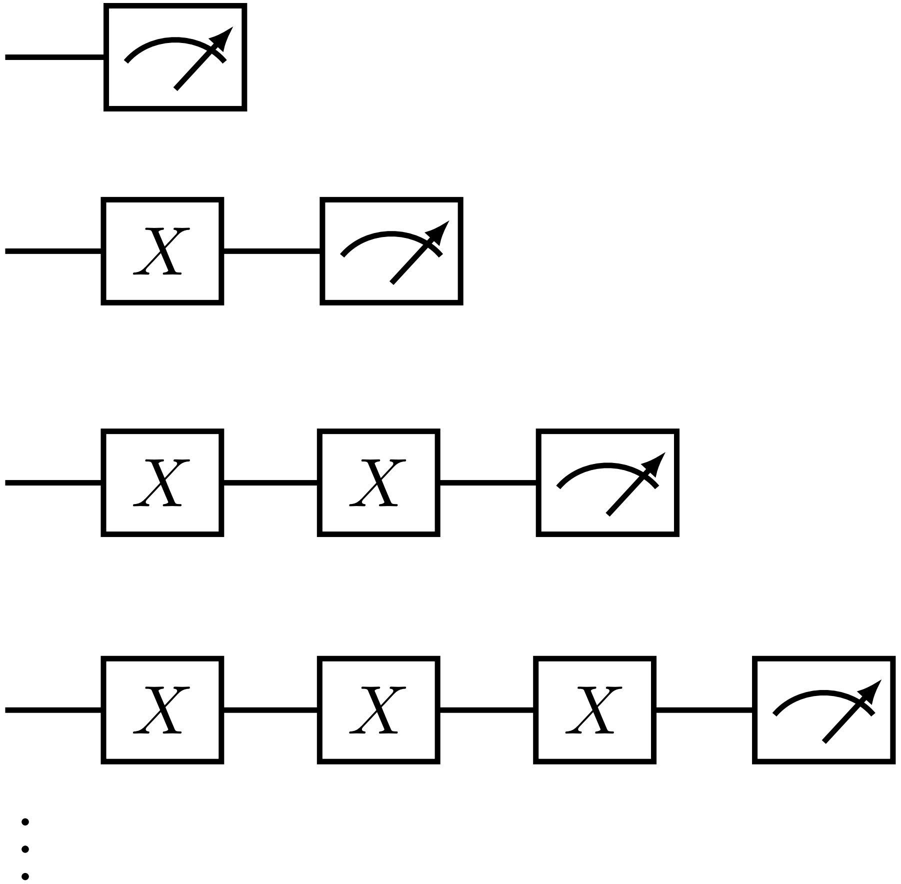
refresher:

$$X\,|0\rangle = |1\rangle$$
$$X\,|1\rangle = |0\rangle$$

$$Z\,|0\rangle = +1\,|0\rangle$$
$$Z\,|1\rangle = -1\,|1\rangle$$

Hello World! to a real quantum experiment

depth



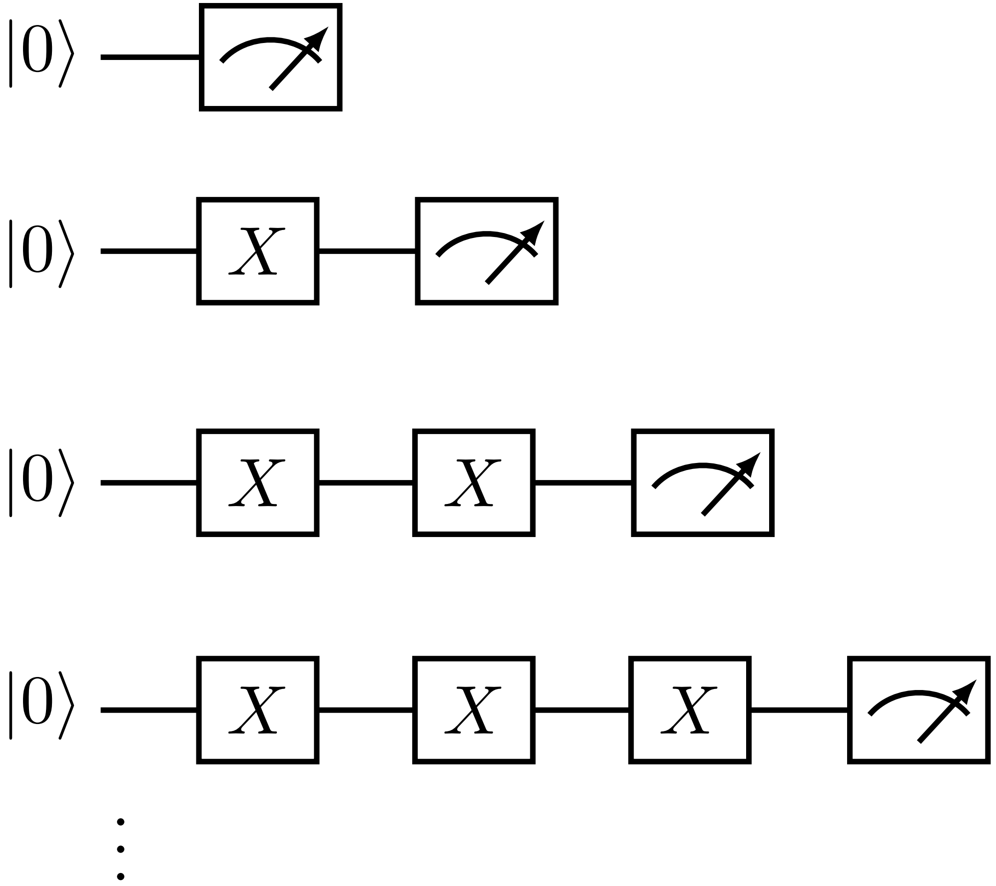
refresher:

$$X|0\rangle = |1\rangle \qquad Z|0\rangle = +1\,|0\rangle$$

$$X|1\rangle = |0\rangle \qquad Z|1\rangle = -1\,|1\rangle$$

Hello World! to a real quantum experiment

depth



refresher:

$$X|0\rangle = |1\rangle \qquad Z|0\rangle = +1|0\rangle$$

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Hello World! to a real quantum experiment

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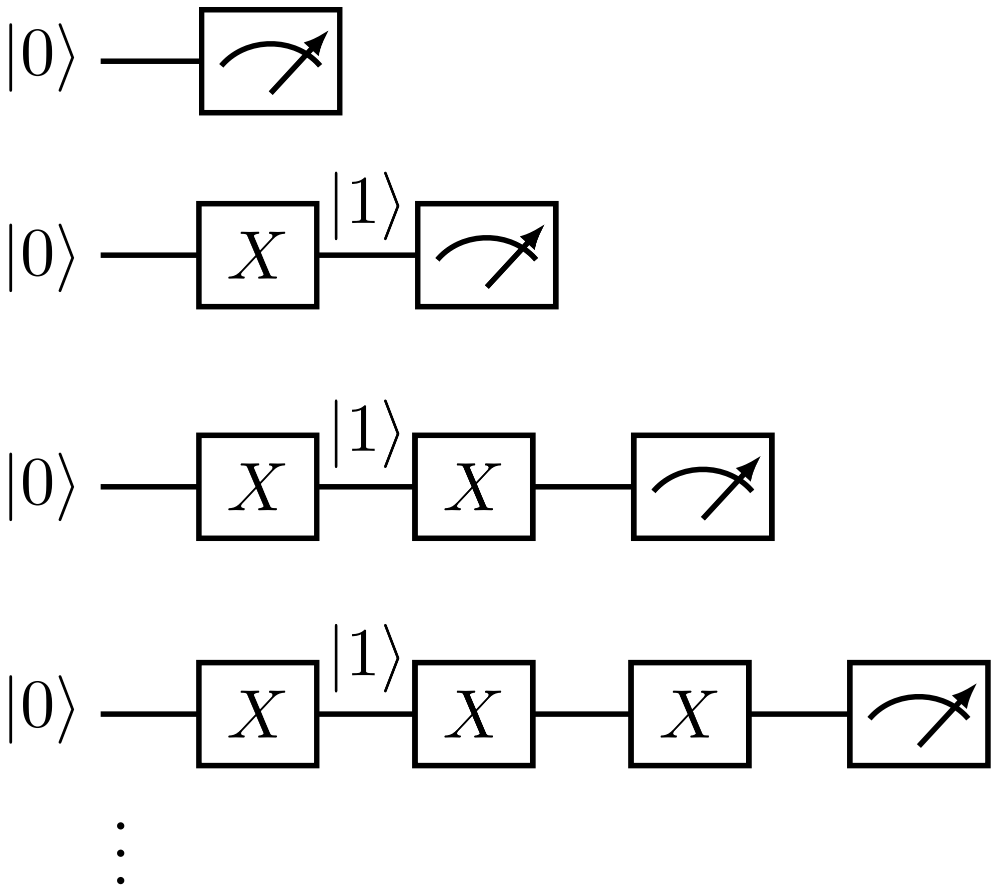

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$$X|0\rangle = |1\rangle \qquad Z|0\rangle = +1|0\rangle$$

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Hello World! to a real quantum experiment

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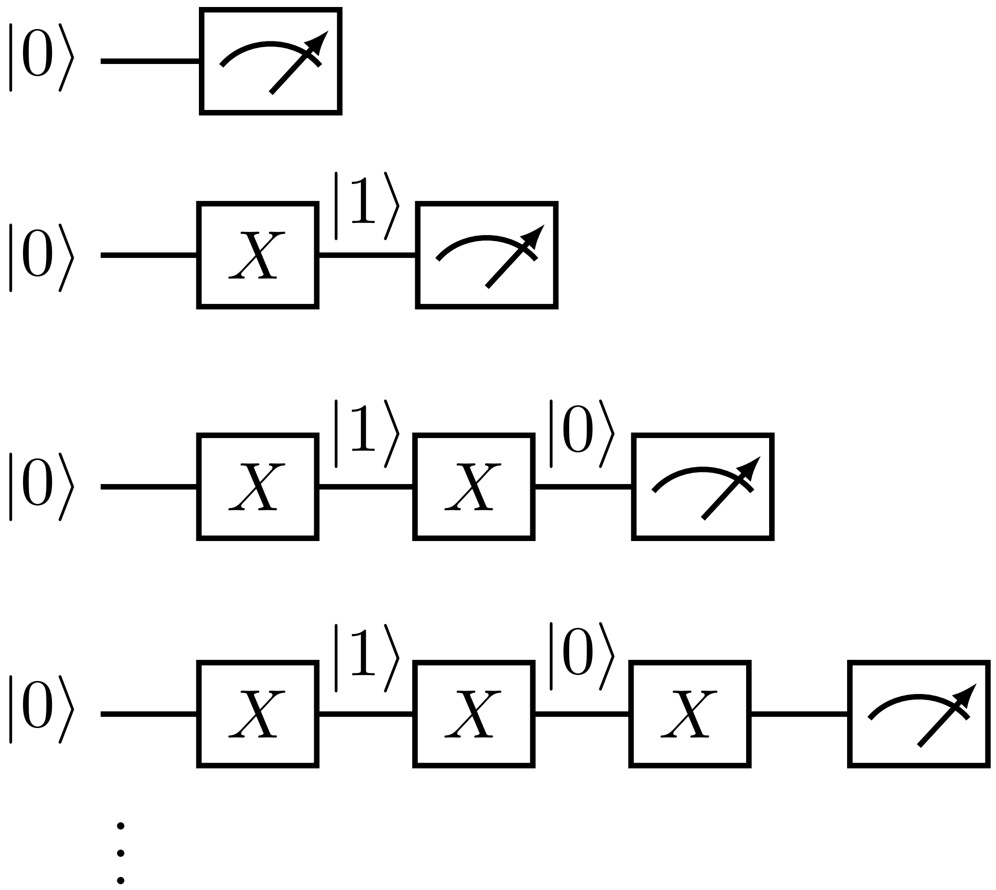

refresher:

$$X|0\rangle = |1\rangle \qquad Z|0\rangle = +1|0\rangle$$

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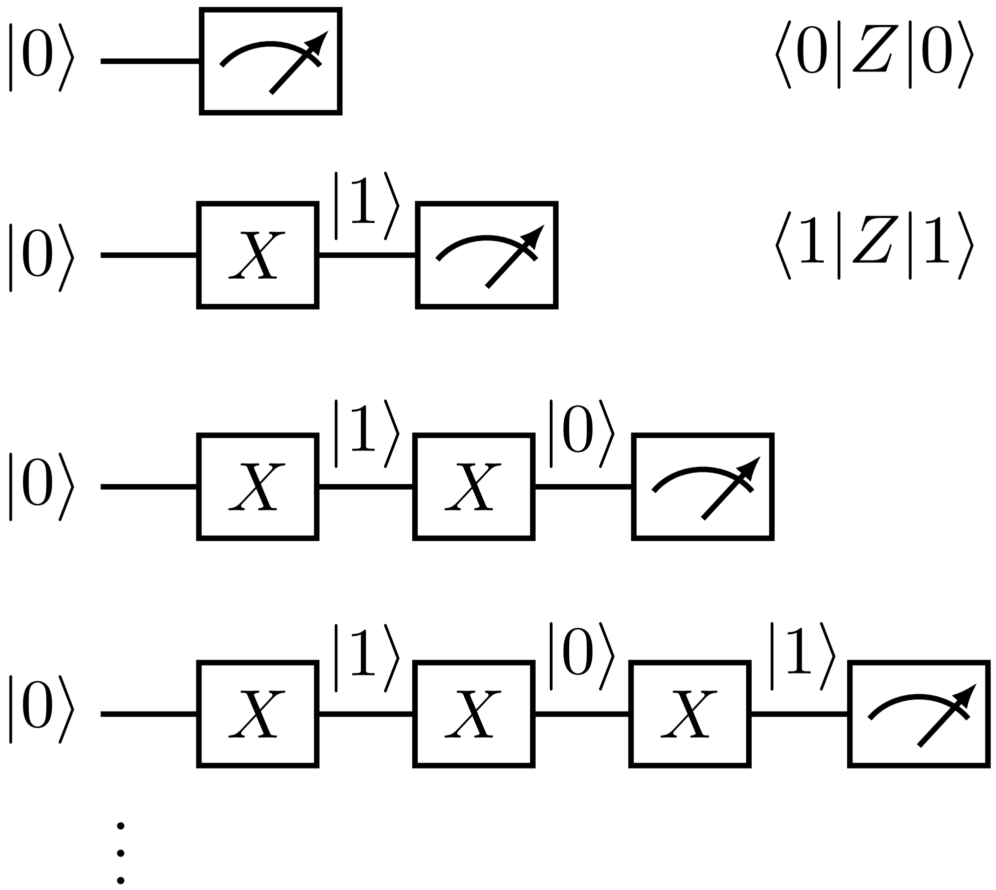
Hello World! to a real quantum experiment

depth



$\langle 0|Z|0\rangle = +1$

$\langle 1|Z|1\rangle = -1$

$$\langle Z\rangle = (-1)^d \,,$$

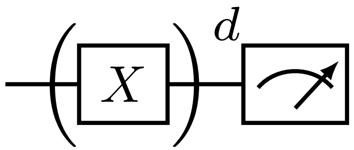
where d is the circuit depth

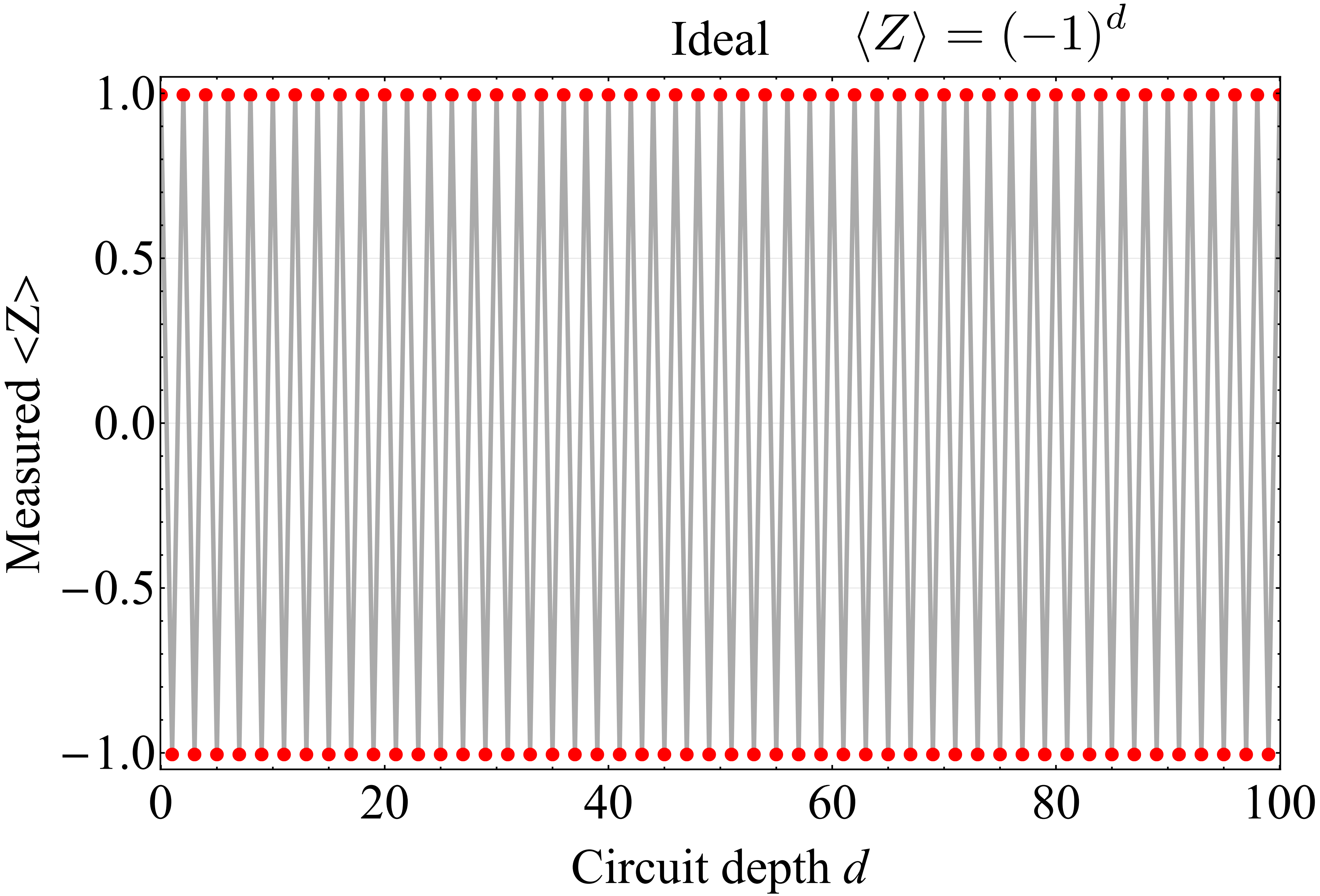
refresher:

$X\,|0\rangle = |1\rangle$ $\qquad Z\,|0\rangle = +1\,|0\rangle$

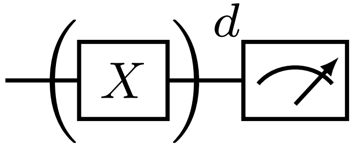
$X\,|1\rangle = |0\rangle$ $\qquad Z\,|1\rangle = -1\,|1\rangle$

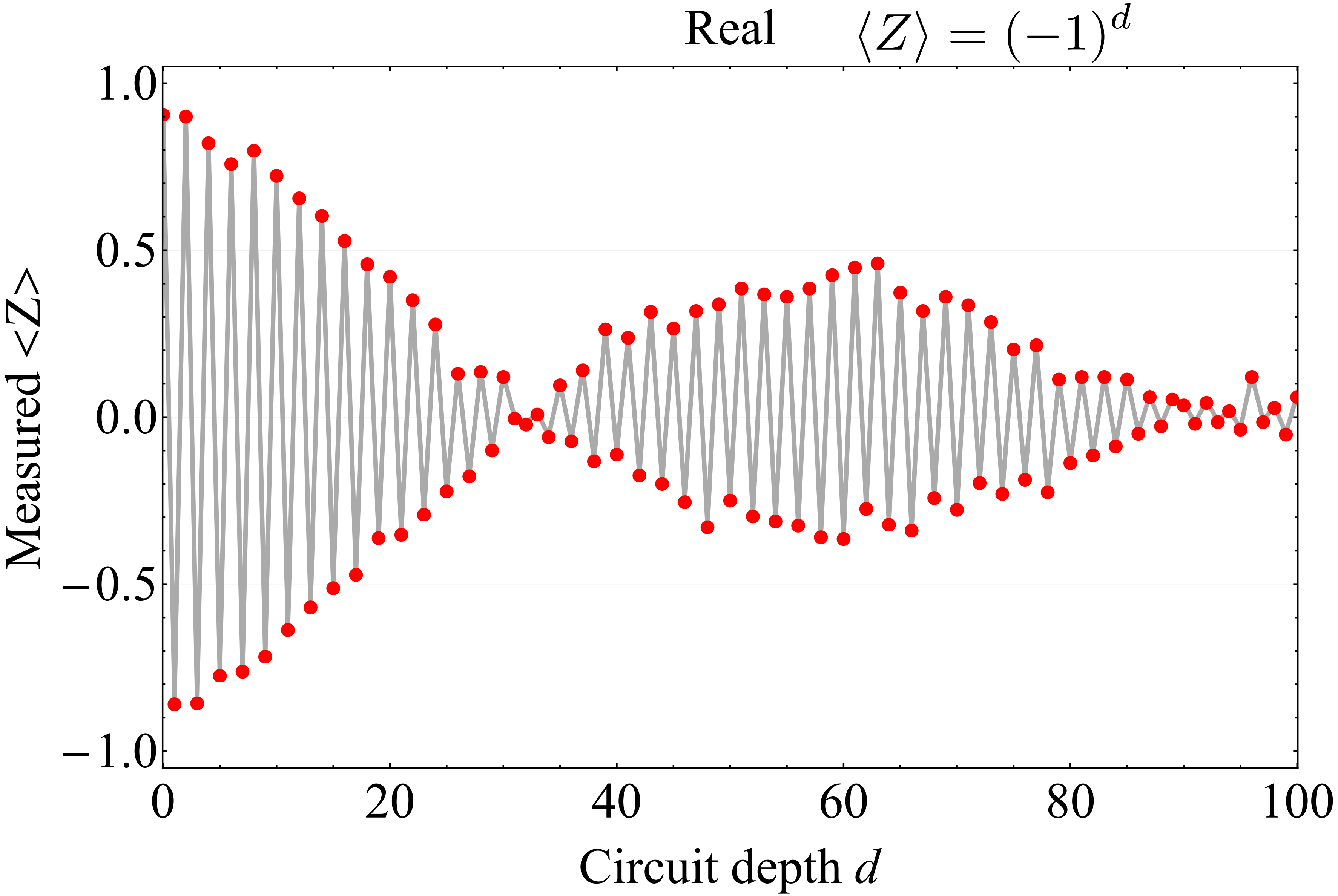
Ideal experiment (warning: doesn't exist)



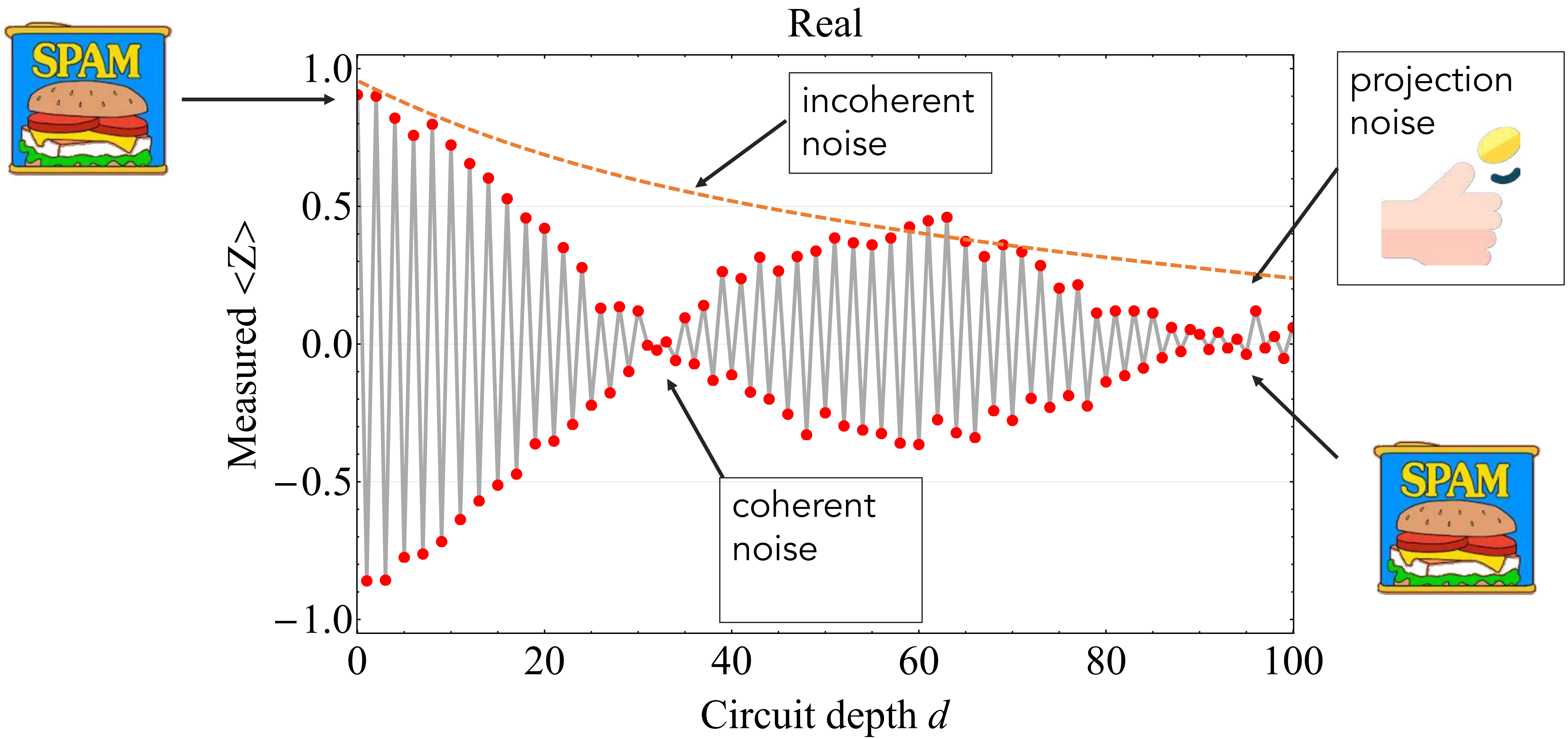


Real experiment





Elements of noise



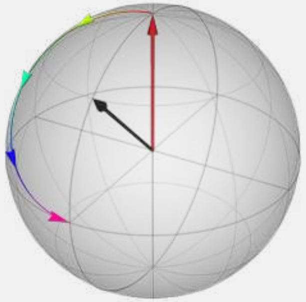
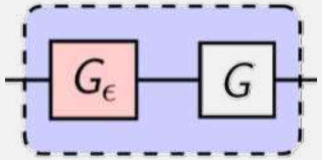
Why study quantum noise?

"Well, your quantum computer is broken in every way possible simultaneously."

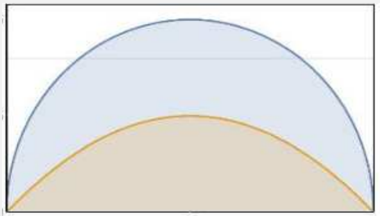
Cartoon: Paul Noth

Our road ahead

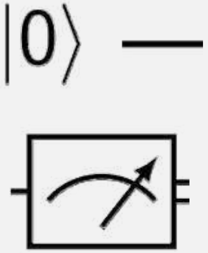
Coherent

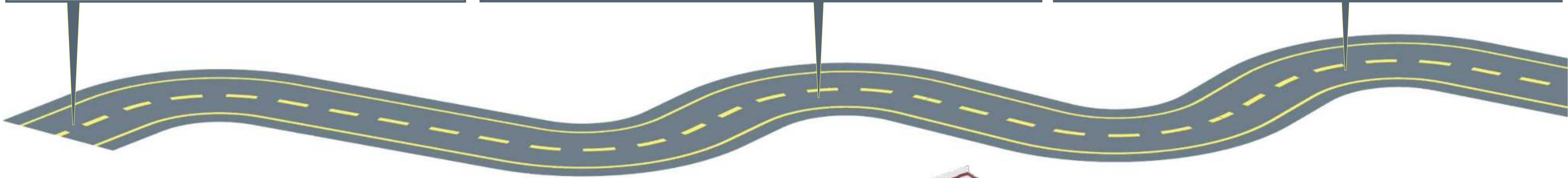



Projection & measurement theory

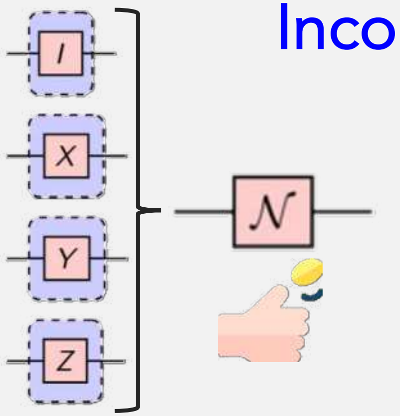



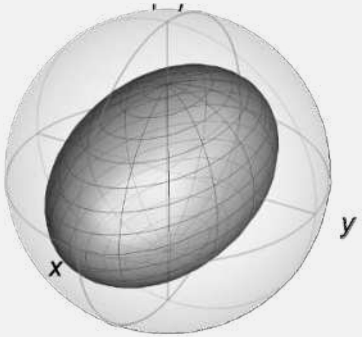
State preparation & measurement





Incoherent

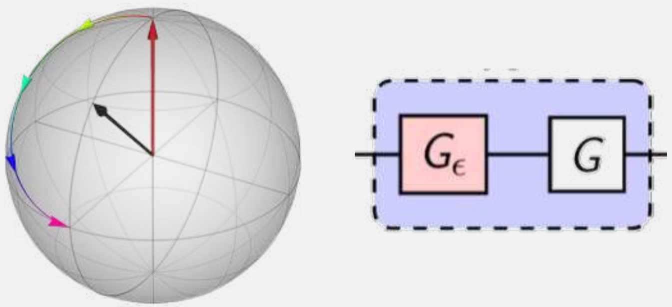


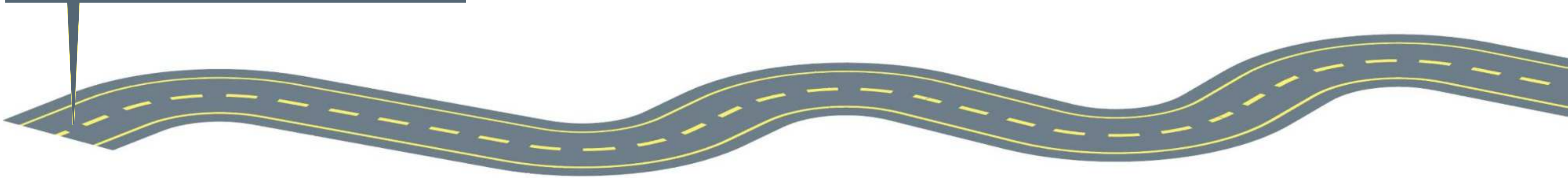


coin toss: flaticon; spam: make it move;
road based on: freepik

Quantum gates meets coherent noise

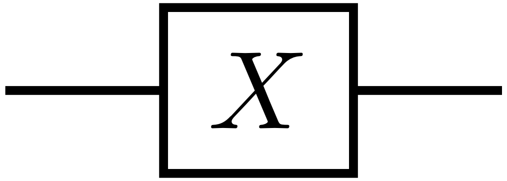
Coherent





road based on: freepik

Gate from time evolution



$$X = R_X(\pi)$$

(up to global phase)

refresher:

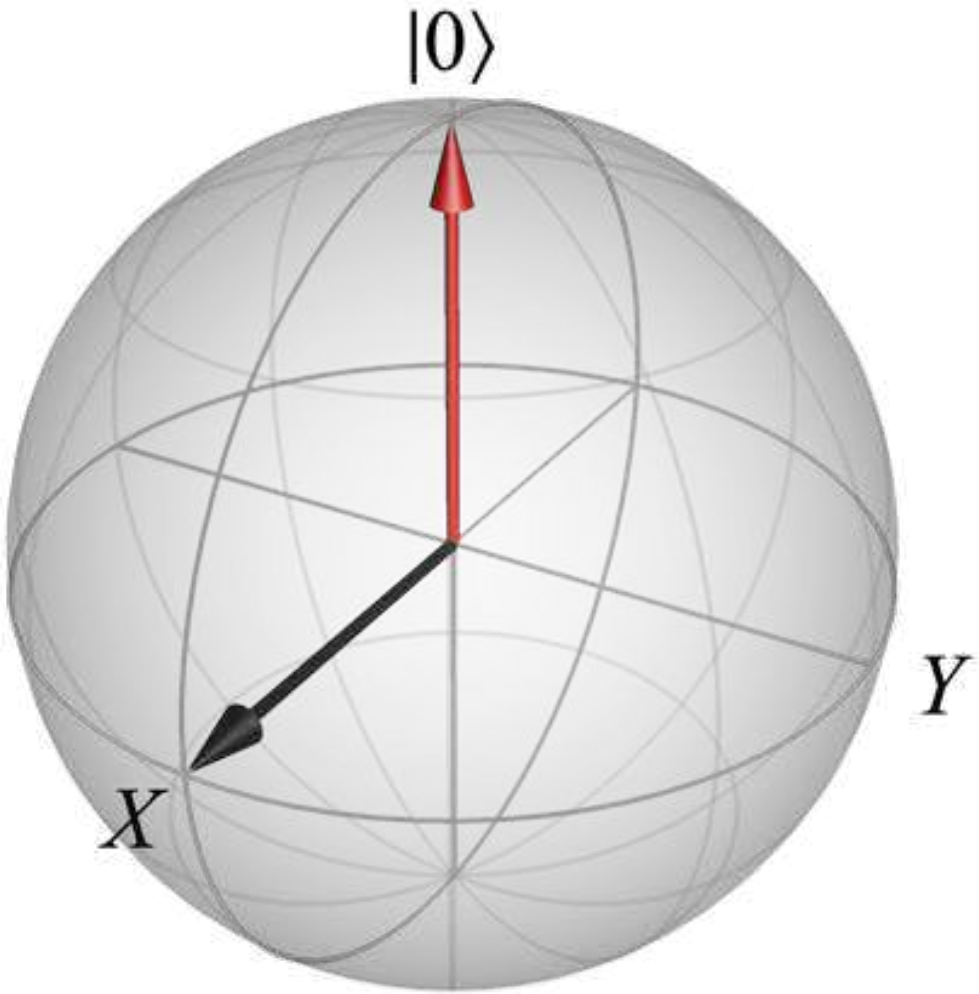
$$\hat{H} = \frac{\hbar\omega}{2} X$$
$$= \frac{\hbar\omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$U(t) = \exp\left(-it\hat{H}/\hbar\right)$$

$$\theta := \omega t$$

$$R_X(\theta) = \exp\left(-\frac{i\theta}{2} X\right)$$
$$= \cos\left(\theta/2\right) I - i \sin\left(\theta/2\right) X$$

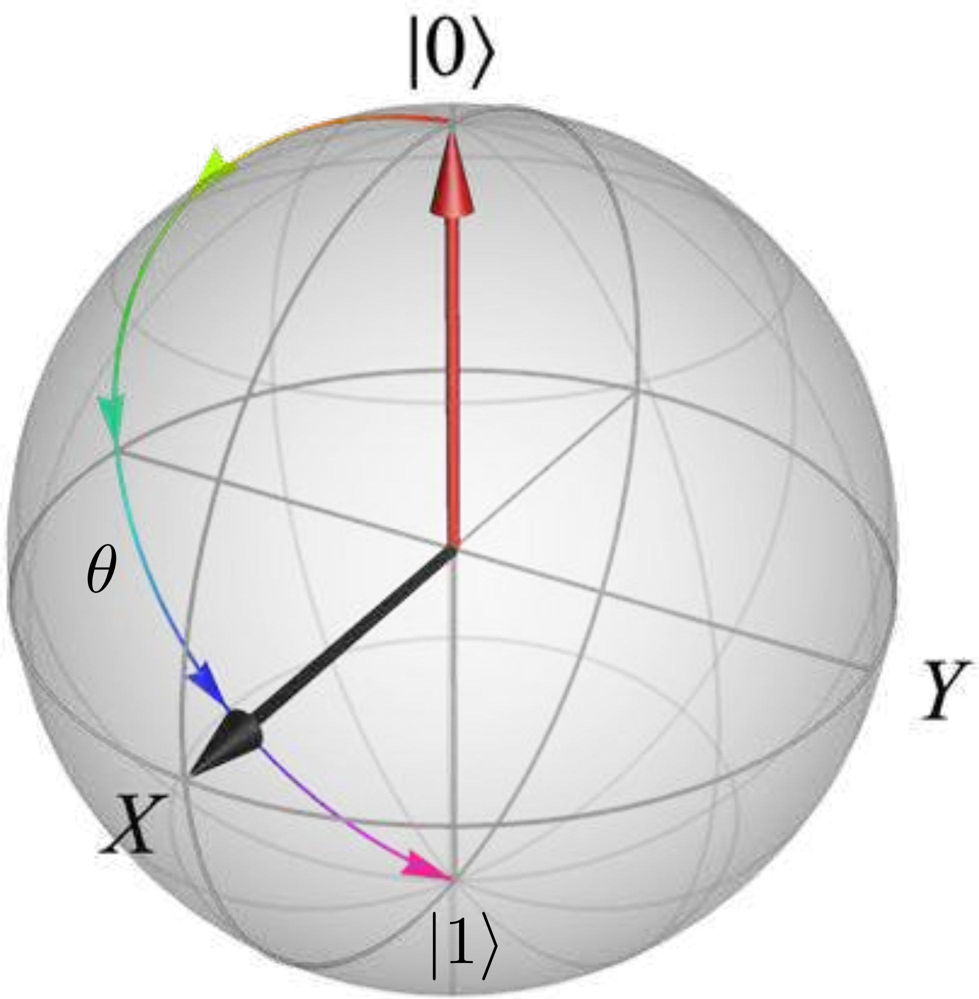
Visualize: Bloch sphere



Visualize: Evolution on the Bloch sphere

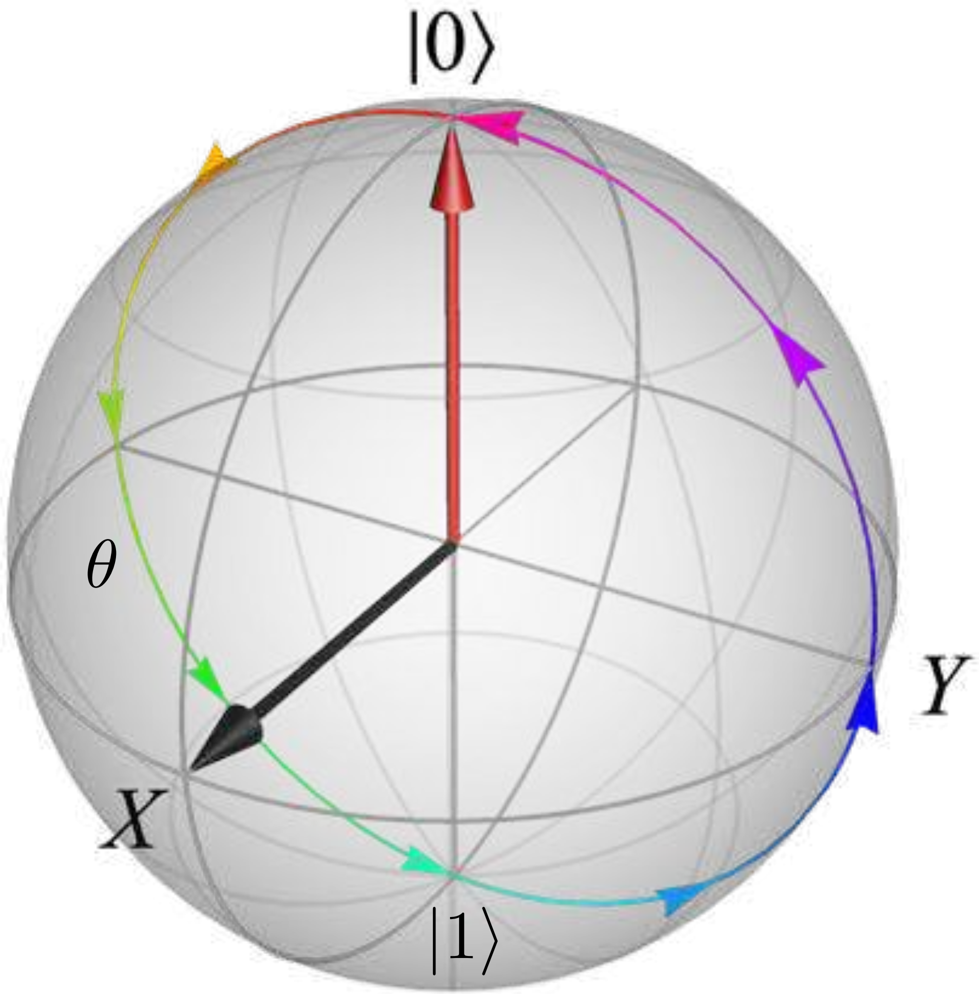
$$R_X(\theta) = \exp\left(-\frac{i\theta}{2}X\right)$$

$$X|0\rangle = |1\rangle$$
$$X|1\rangle = |0\rangle$$



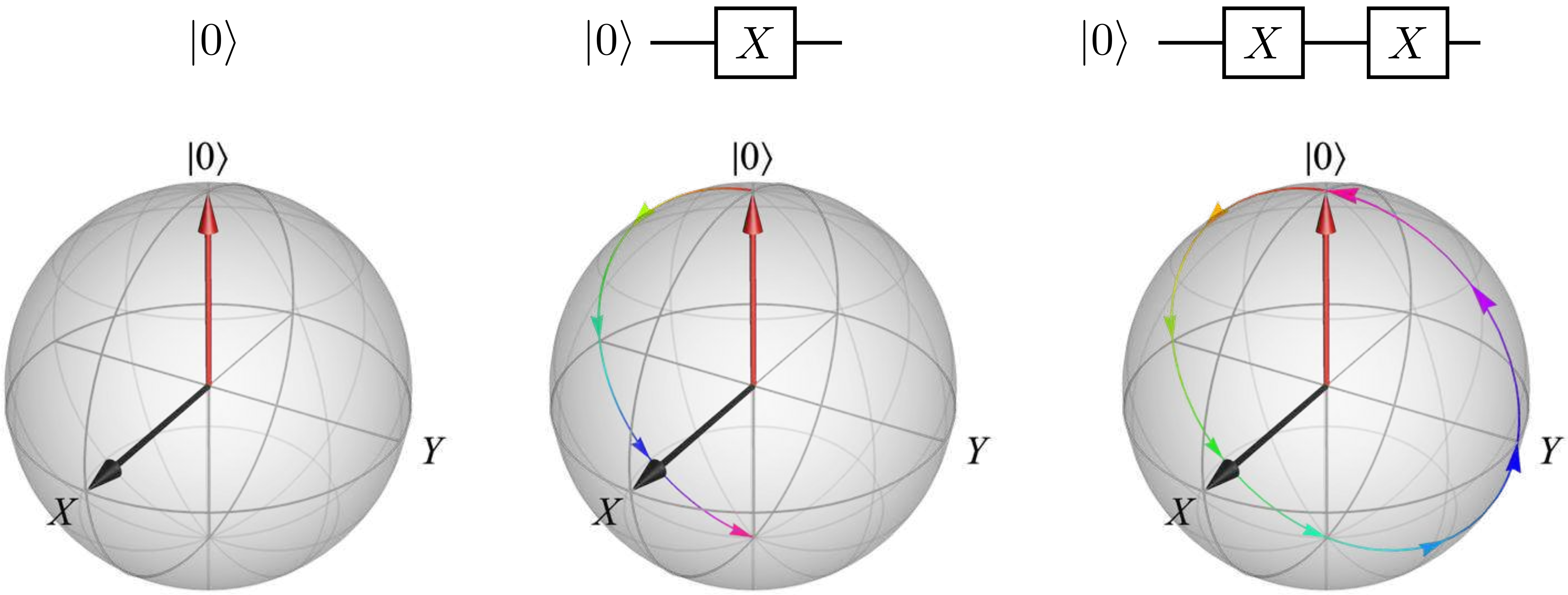
Visualize: Evolution on the Bloch sphere

$$R_X(\theta) = \exp\left(-\frac{i\theta}{2}X\right)$$



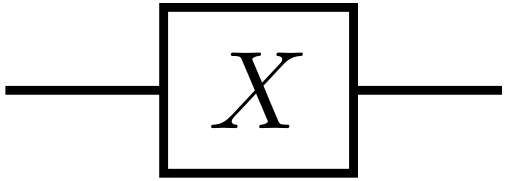
$$X\,|0\rangle = |1\rangle$$
$$X\,|1\rangle = |0\rangle$$

Evolution on the Bloch sphere

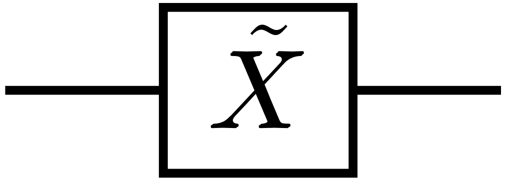


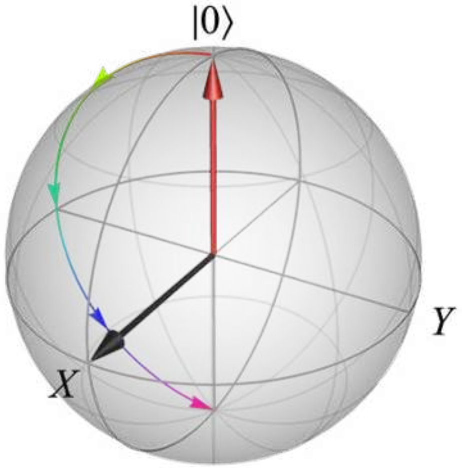
Miscalibrated gate

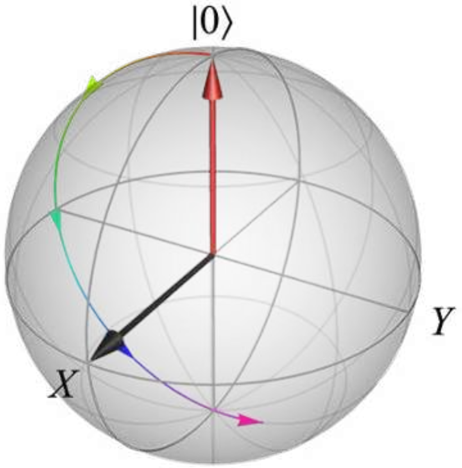
Ideal gate



Noisy gate







Show

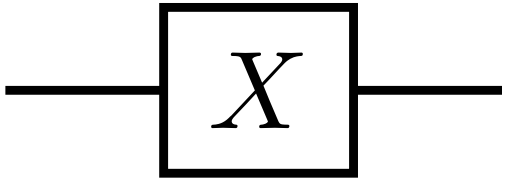
$$R_X(\theta + \phi) = R_X(\theta)\, R_X(\phi)$$

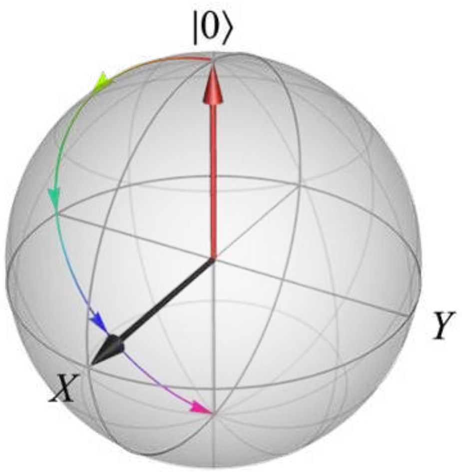
$$X = R_X(\pi)$$

$$\tilde{X} := R_X(\pi + \epsilon)$$

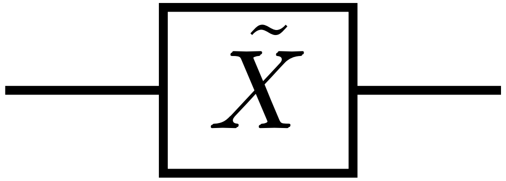
Miscalibrated gate

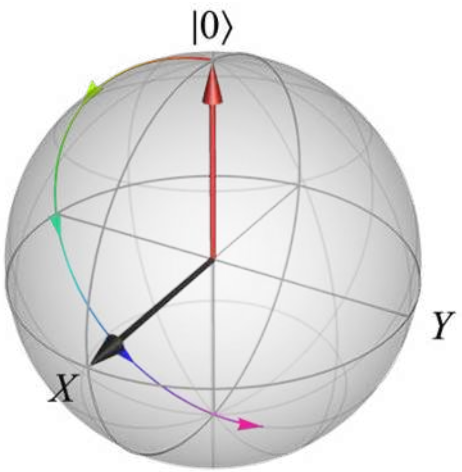
Ideal gate





Noisy gate





Show

$$R_X(\theta+\phi) = R_X(\theta)\, R_X(\phi)$$

$$X = R_X(\pi)$$

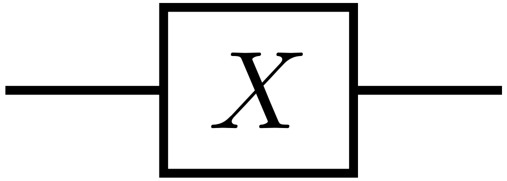
$$\tilde{X} := R_X(\pi+\epsilon)$$

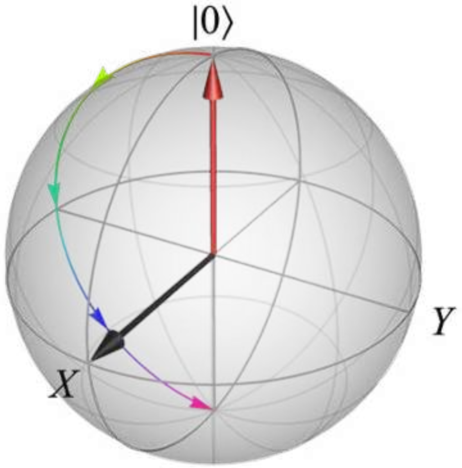
$$R_X(\theta) = \exp\left(-\frac{i\theta}{2}X\right)$$
$$= \cos(\theta/2) I - i\sin(\theta/2) X$$

$$= \exp\left(-i\frac{\pi+\epsilon}{2}X\right)$$
$$= \exp\left(-i\frac{\pi}{2}X - i\frac{\epsilon}{2}X\right)$$
$$= \exp\left(-i\frac{\pi}{2}X\right)\exp\left(-i\frac{\pi}{2}X\right)$$
$$= R_X(\epsilon)\, R_X(\pi)$$
$$= X_\epsilon X$$

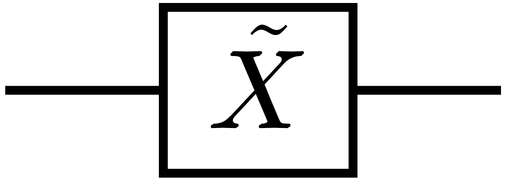
Noisy gate decomposition

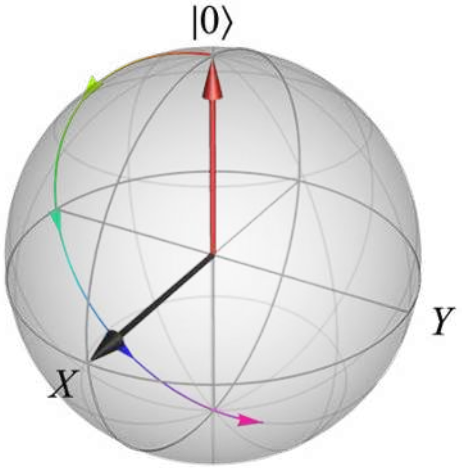
Ideal gate

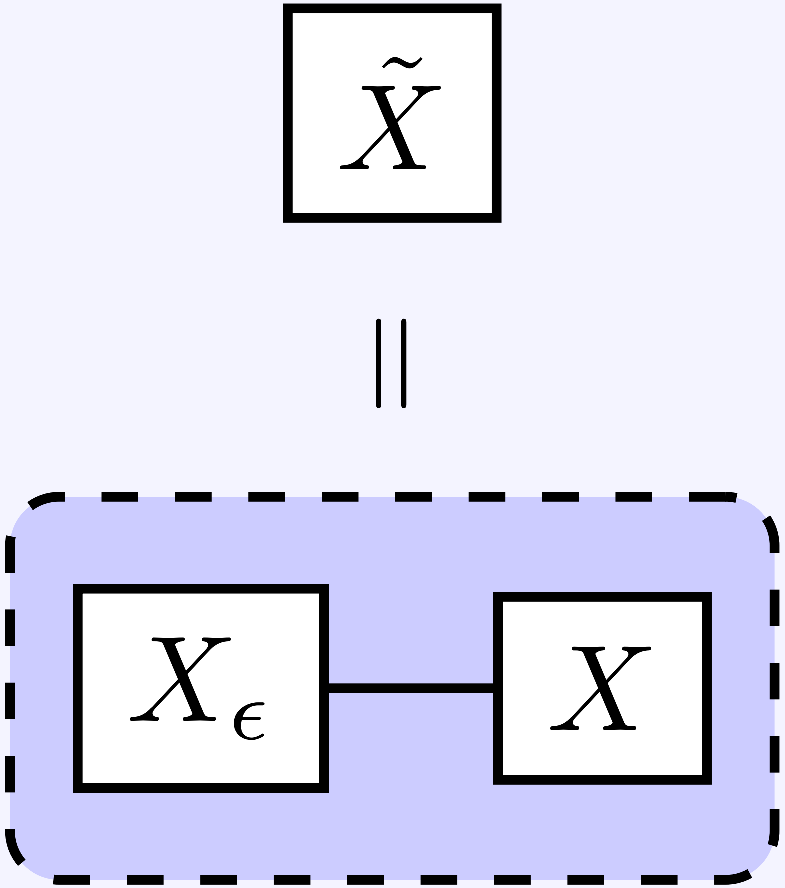




Noisy gate

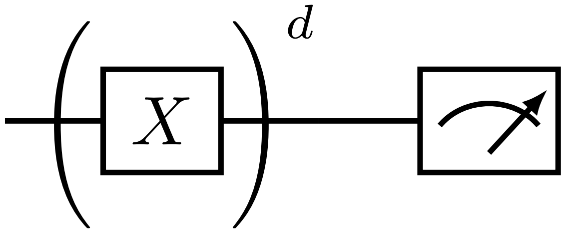






Using a noisy gate in a quantum circuit

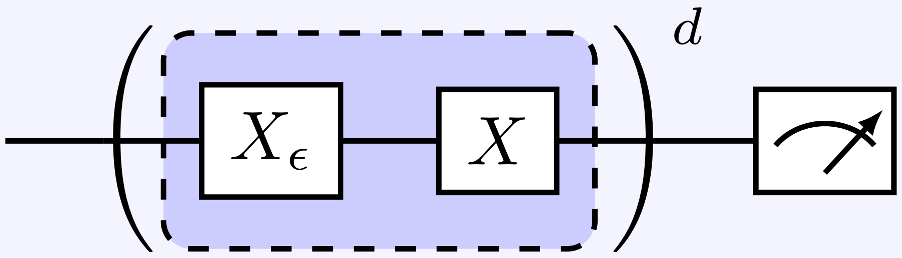
Ideal



$$X = R_X(\pi)$$

$$\begin{aligned} U_{\text{total}} &= X^d \\ &= \left[R_X\left(\pi\right)\right]^d \\ &= R_X\left(d\pi\right) \end{aligned}$$

Noisy



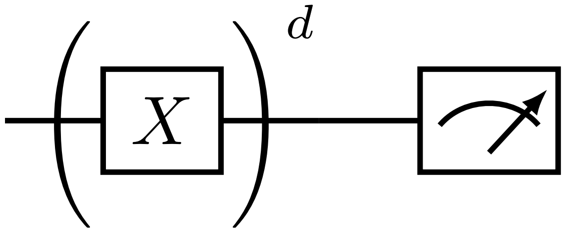
$$\tilde{X} := R_X\left(\pi + \epsilon\right) = X_\epsilon X$$

$$\begin{aligned} \tilde{U}_{\text{total}} &= \tilde{X}^d \\ &= \left[R_X\left(\epsilon\right) R_X\left(\pi\right)\right]^d \\ &= R_X\left(d\epsilon\right) R_X\left(d\pi\right) \end{aligned}$$

$$\tilde{U}_{\text{total}} = R_X\left(d\epsilon\right) U_{\text{total}}$$

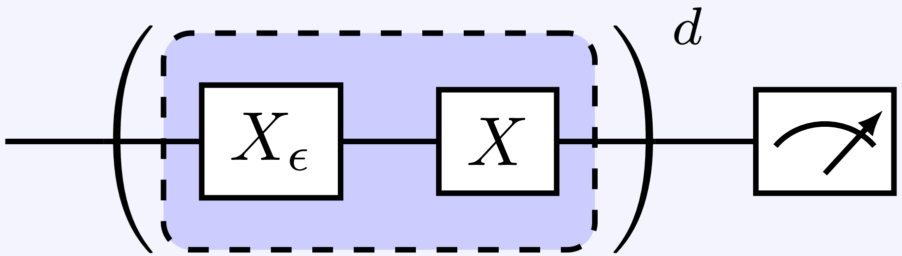
Using a noisy gate in a quantum circuit: final state

Ideal



$$U_{\text{total}} = X^d = R_X\left(d\pi\right)$$

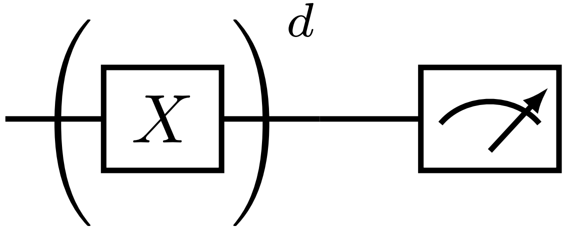
Noisy



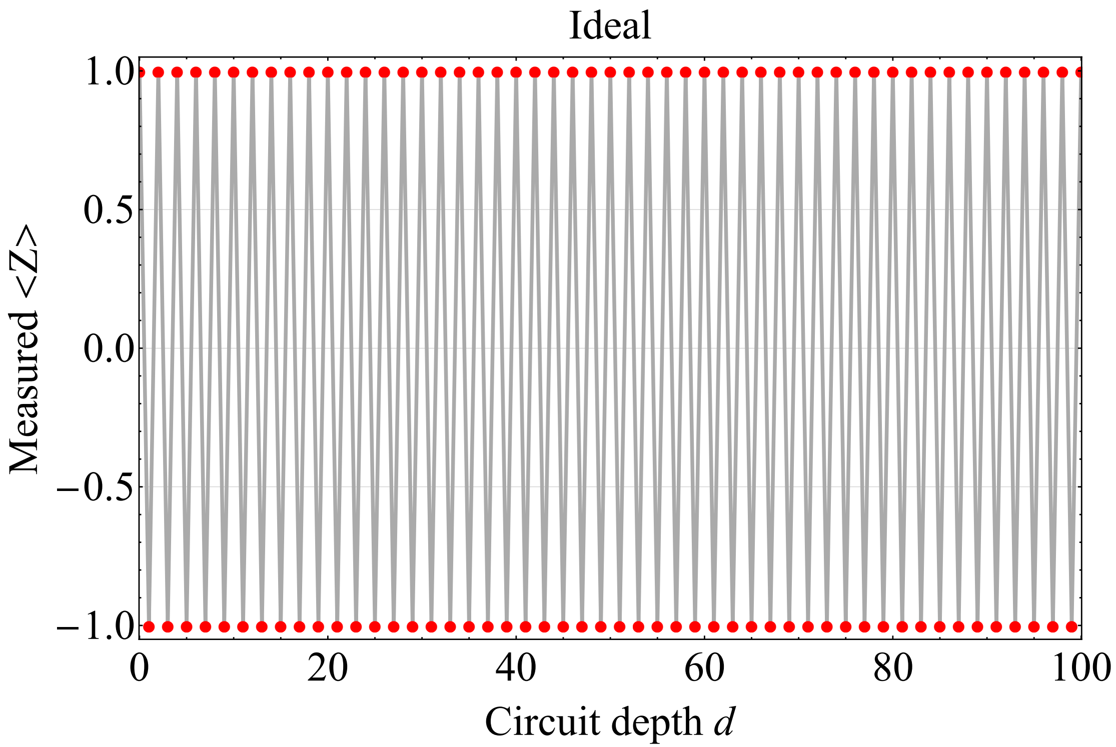
$$\tilde{U}_{\text{total}} = R_X\left(d\epsilon\right) U_{\text{total}}$$

Ideal vs. noisy observable

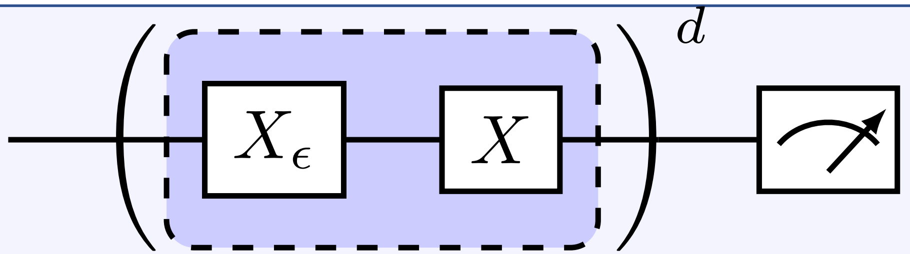
Ideal



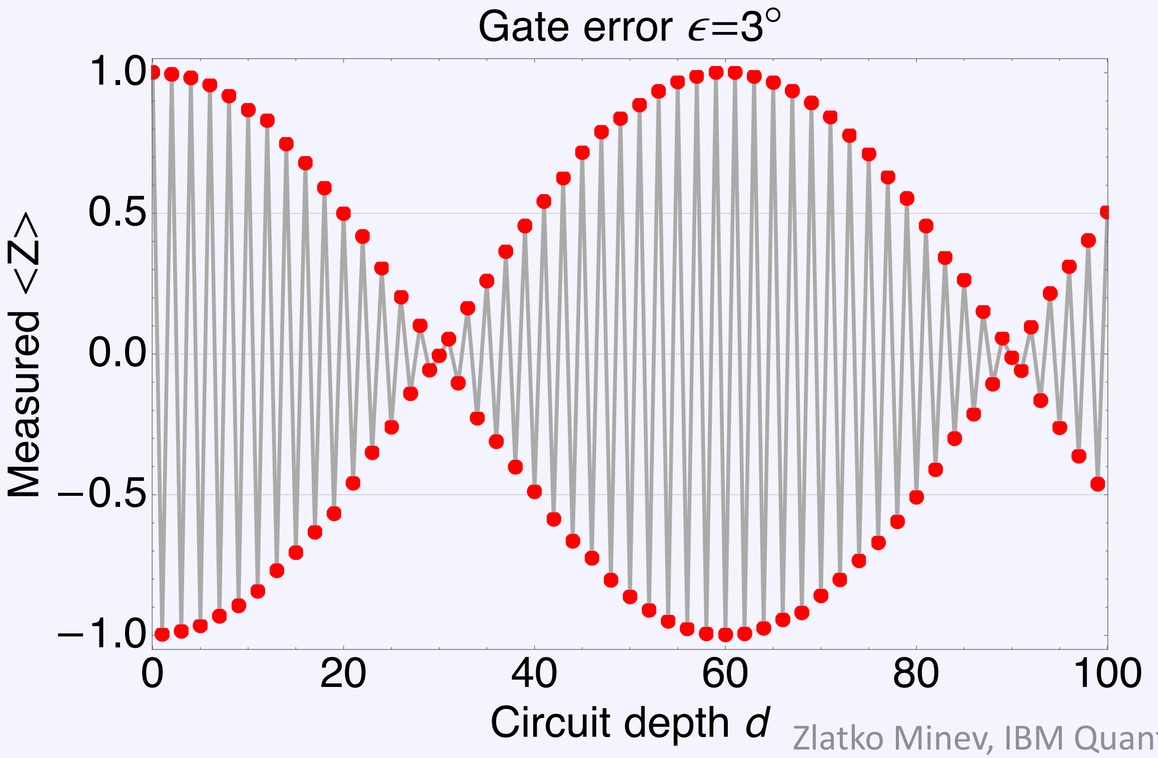
$$\langle \psi_f | Z | \psi_f \rangle = \cos(d\pi) = (-1)^d$$



Noisy

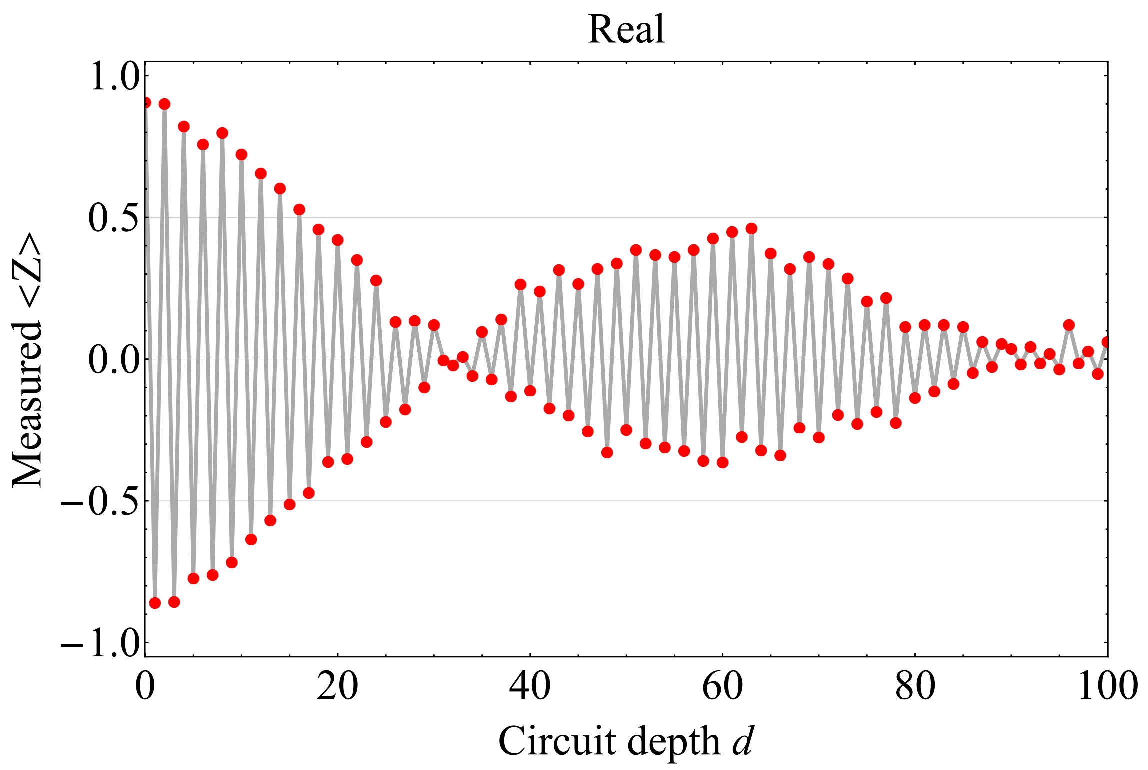


$$\left\langle \tilde{\psi}_f \middle| Z \middle| \tilde{\psi}_f \right\rangle = \cos(d\pi + d\epsilon)$$

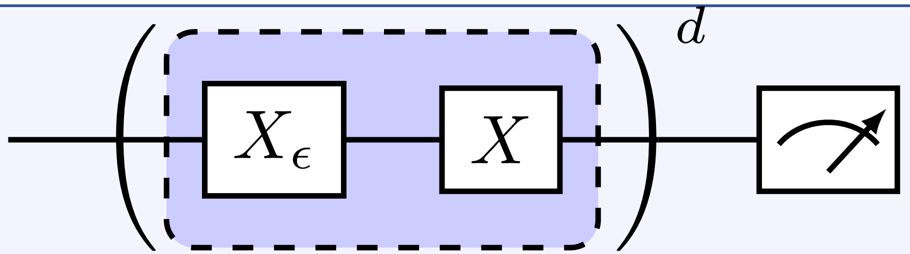


Compare to full experiment

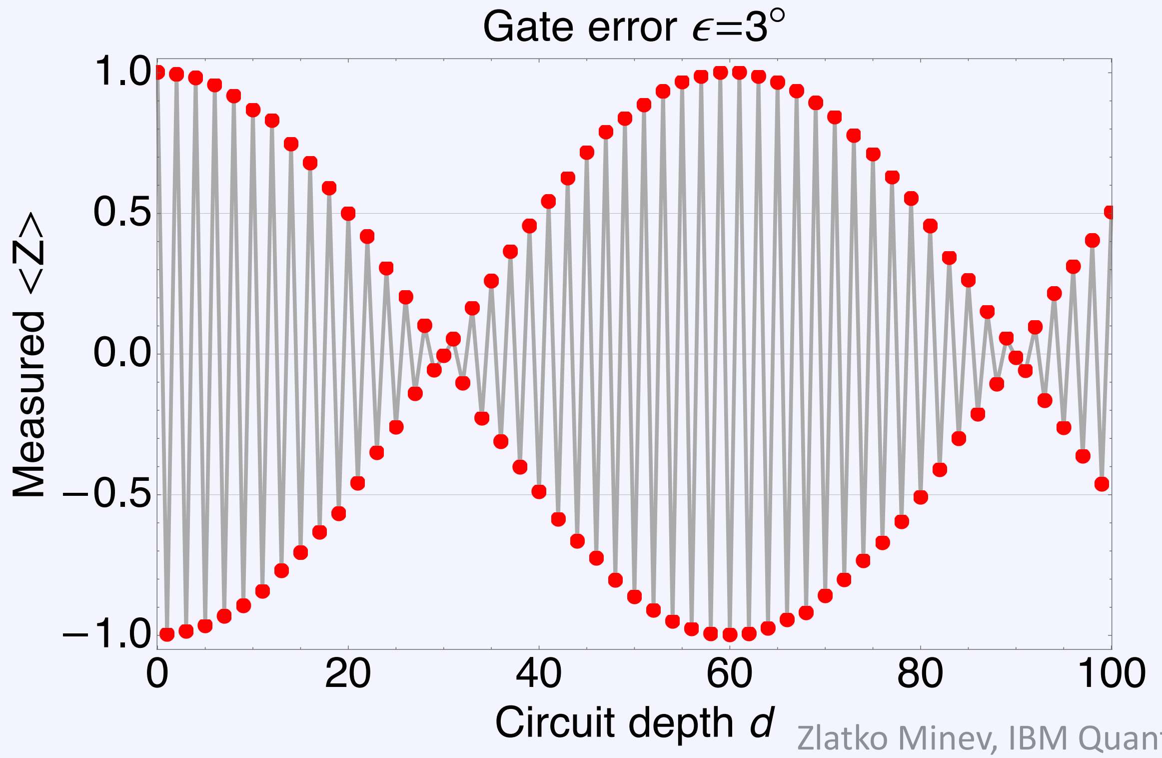
Full experiment



Noisy



$$\left\langle \tilde{\psi}_f \middle| Z \middle| \tilde{\psi}_f \right\rangle = \cos\left(d\pi + d\epsilon\right)$$



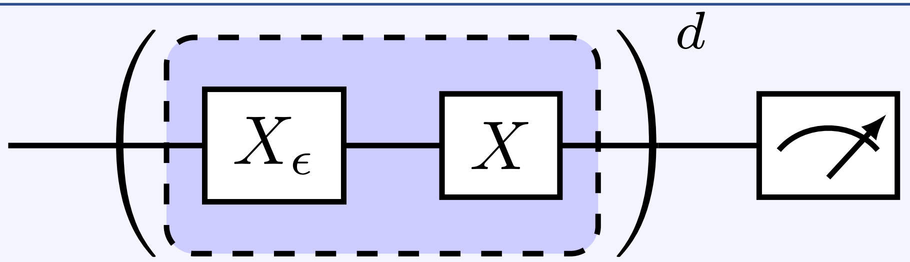
Coherent error is bad, quadratically so

Coherent errors have a *quadratic* impact on algorithmic accuracy (worst-case error)

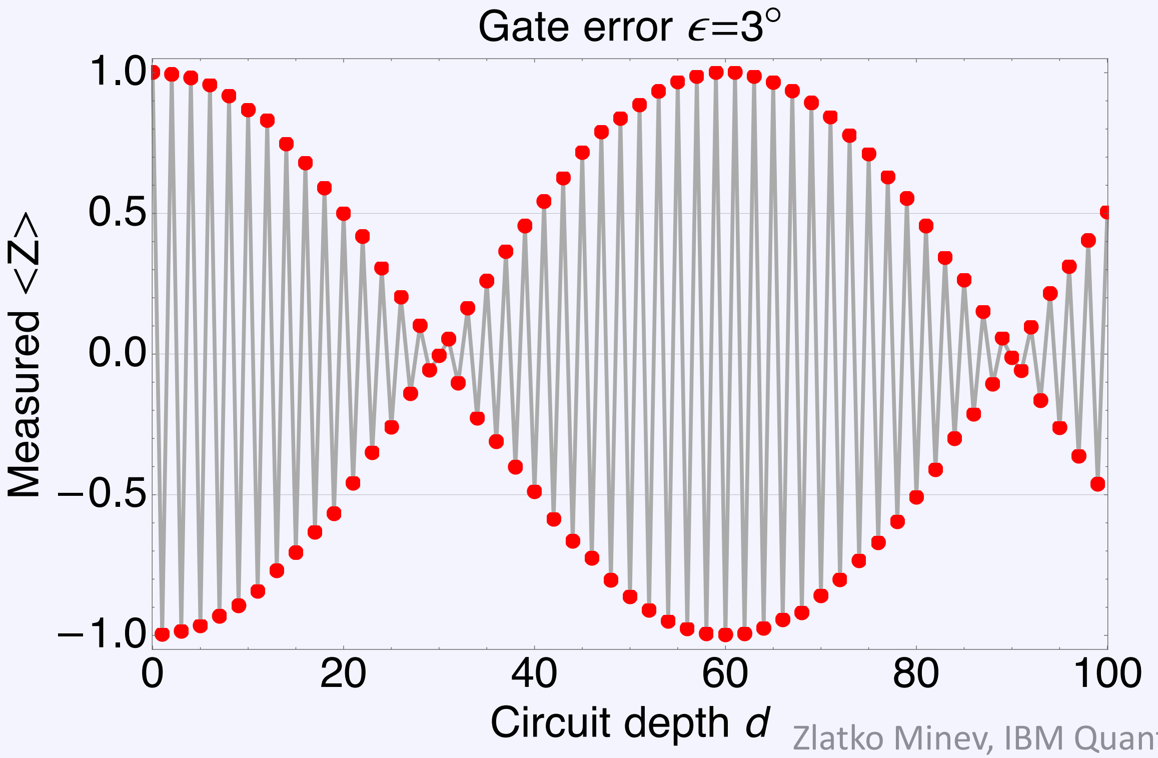
$$\left\langle \tilde{\psi}_f \middle| Z \middle| \tilde{\psi}_f \right\rangle - \langle \psi_f | Z | \psi_f \rangle$$

$$\cos(x) = 1 - \frac{1}{2}x^2 + \mathcal{O}\left(x^3\right)$$

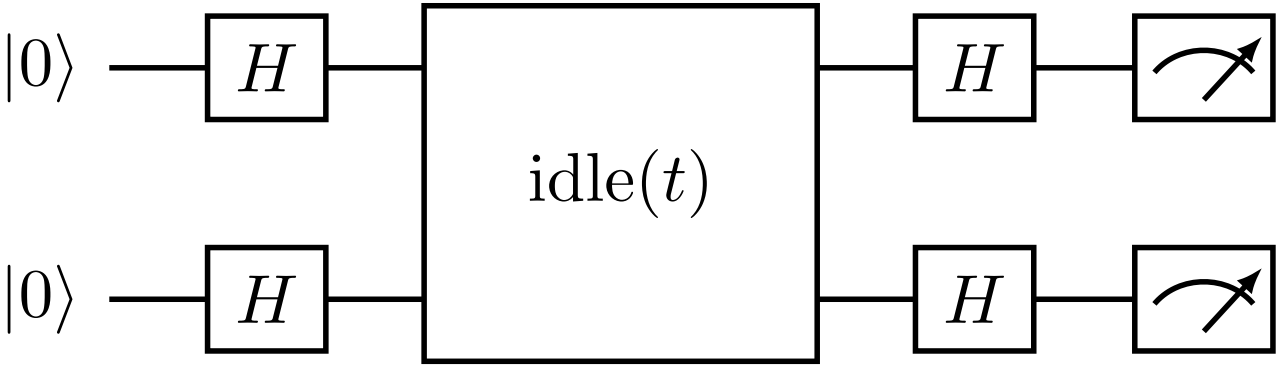
Noisy



$$\left\langle \tilde{\psi}_f \middle| Z \middle| \tilde{\psi}_f \right\rangle = \cos\left(d\pi + d\epsilon\right)$$



Example: two-qubit ZZ error



Breakout to notebook

Hadamard gate

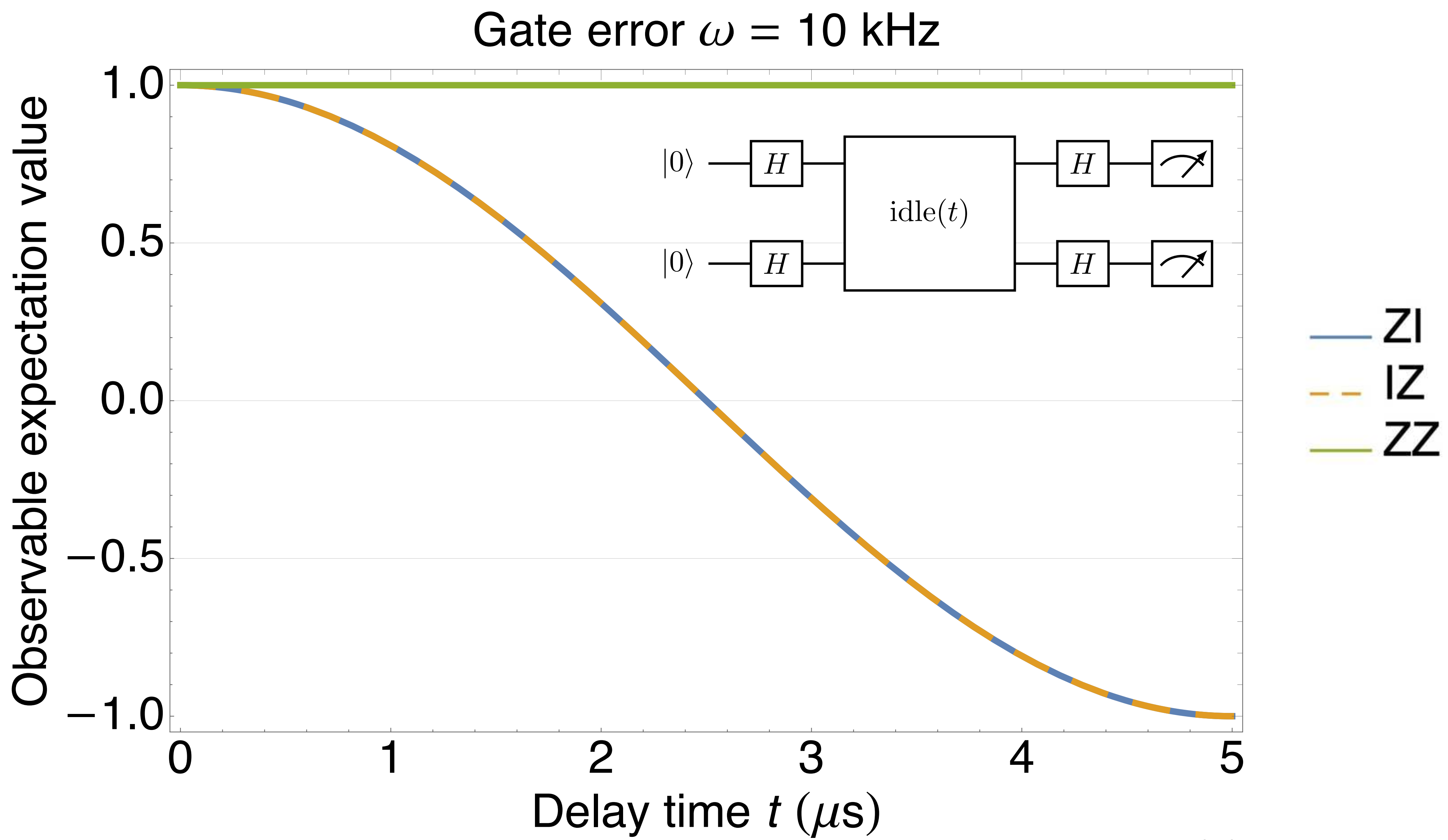
$$\boxed{H}$$

$$H = \begin{array}{c c} & \begin{array}{cc} \langle 0| & \langle 1| \end{array} \\ \begin{array}{c} |0\rangle \\ |1\rangle \end{array} & \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{pmatrix} \end{array}$$

$$H|0\rangle = |+x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$H|1\rangle = |-x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

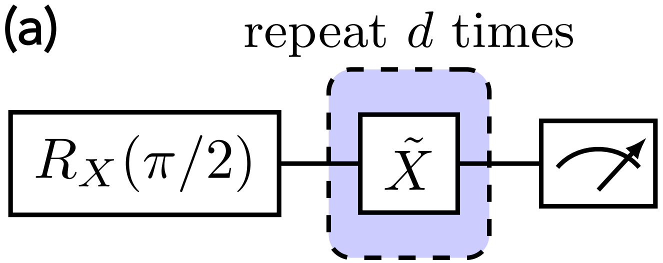
Example: two-qubit ZZ error



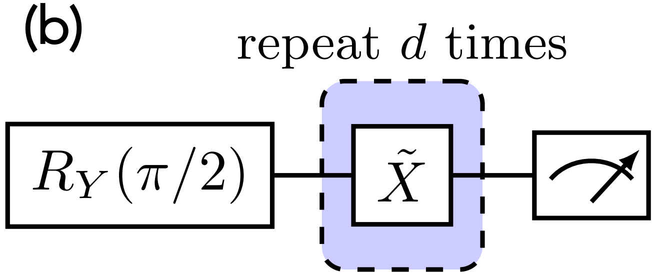
Dive deeper? Try the following

1. Amplitude-error amplification sequences.
   (A) Calculate the expectation value <*Z*(*d*)> as a function of depth *d* of the sequence for the following sequences. Assume the noise model $\tilde{X} := R_X(\pi + \epsilon) = X_\epsilon X$
   (B) How could you use this result to fine-tune your gates?
   (C) Can you come up with an alternative or more clever error-amplification sequence?

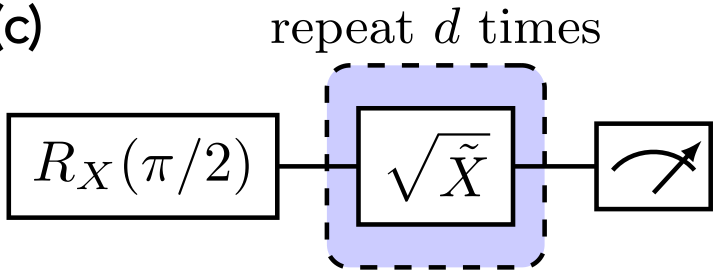
(a)



(b)

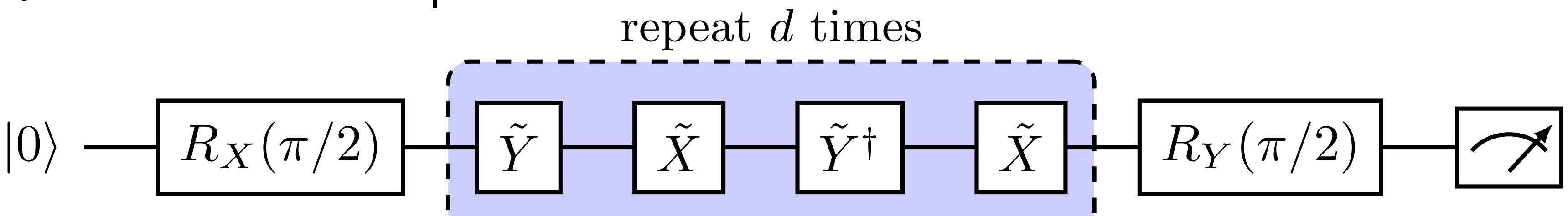


(c)



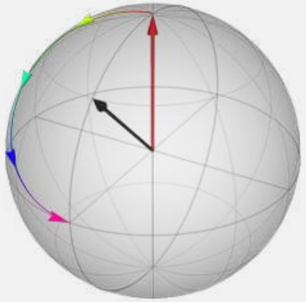
2. Phase-error amplification sequences. Use the noise model: Repeat (A), (B), and (C) for Exercise (1) for the following sequences and assuming a phase error between the X and Y gates, rather than an amplitude error.

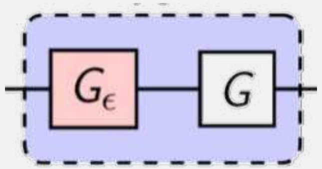
$$\tilde{X} = X\,,$$
$$\tilde{Y} = \exp\left[-i\frac{\pi}{2}\left(\cos\epsilon\, Y + \sin\epsilon\, X\right)\right]\,.$$



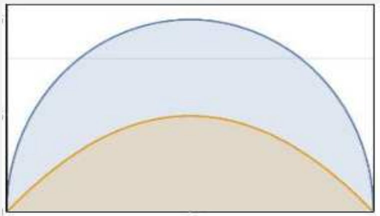
Our road ahead

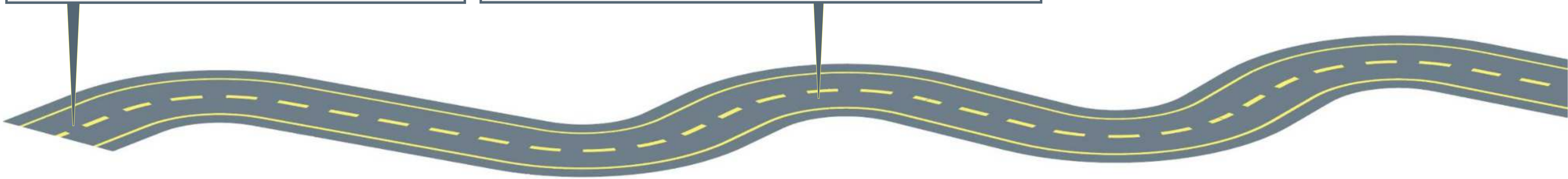
Coherent





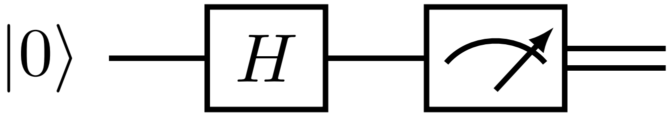
Projection & measurement theory

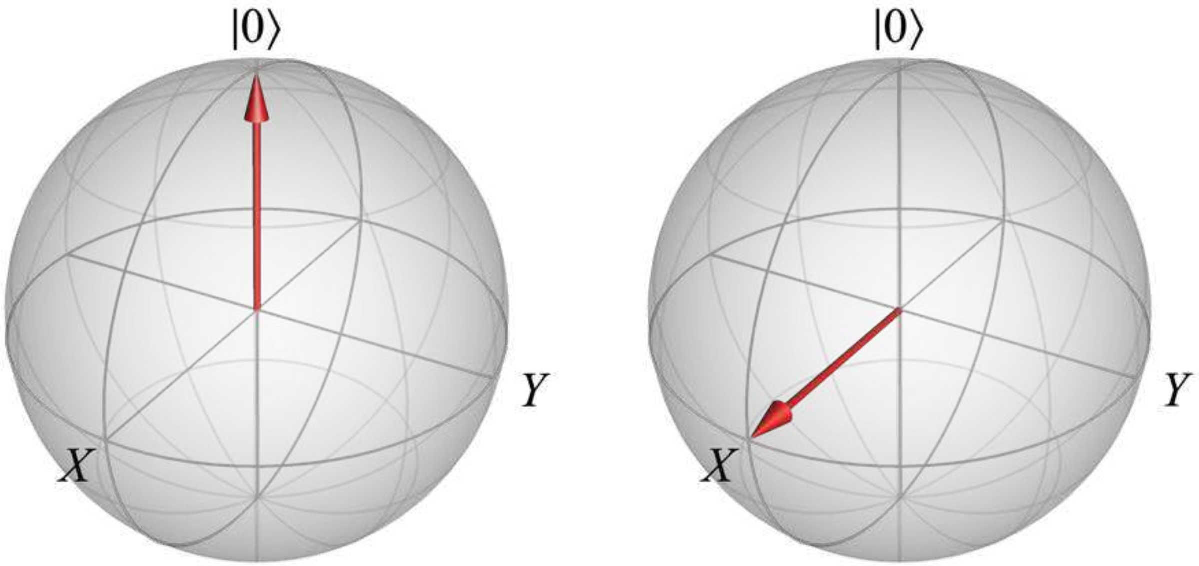




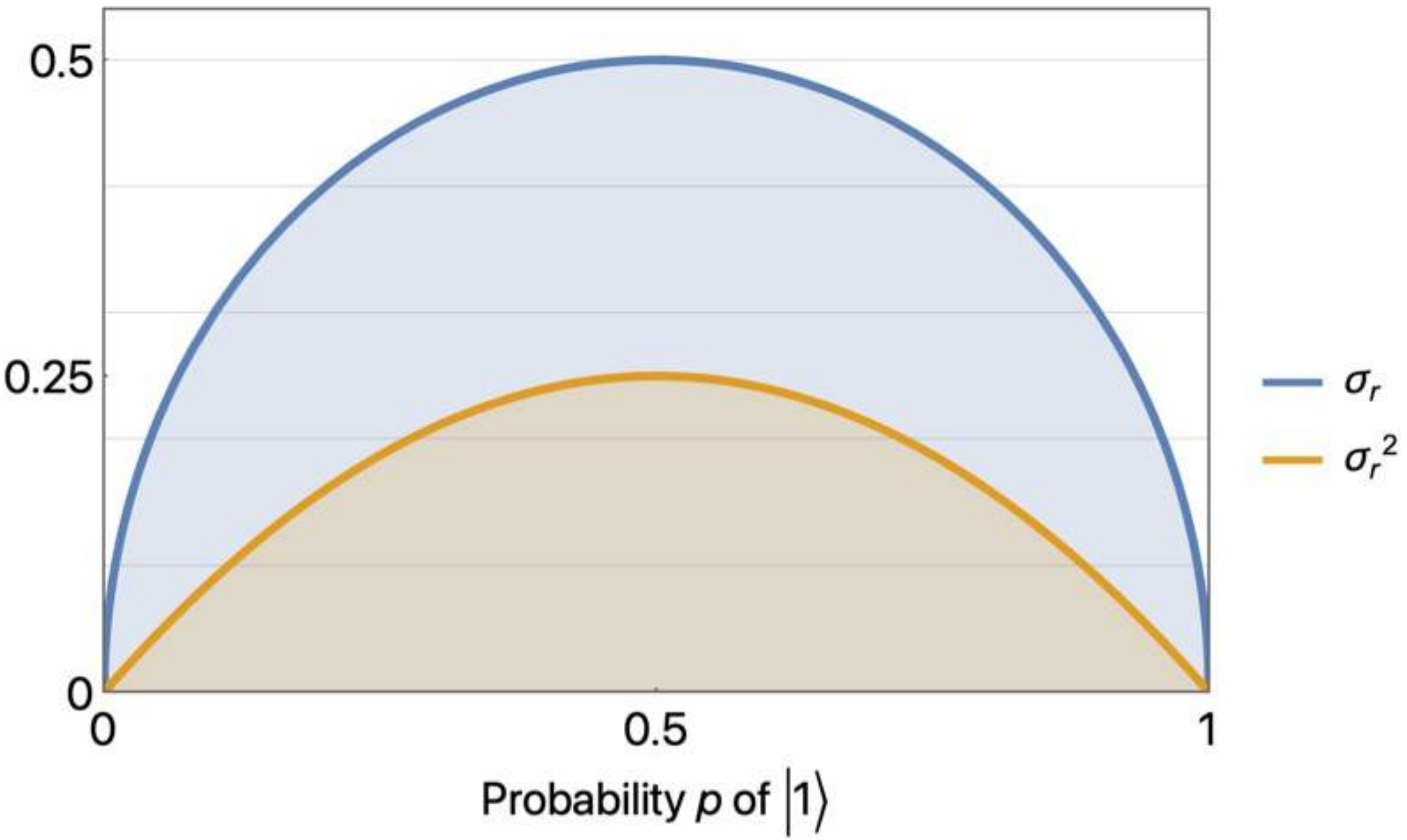
coin toss: flaticon; spam: make it move;
road based on: freepik

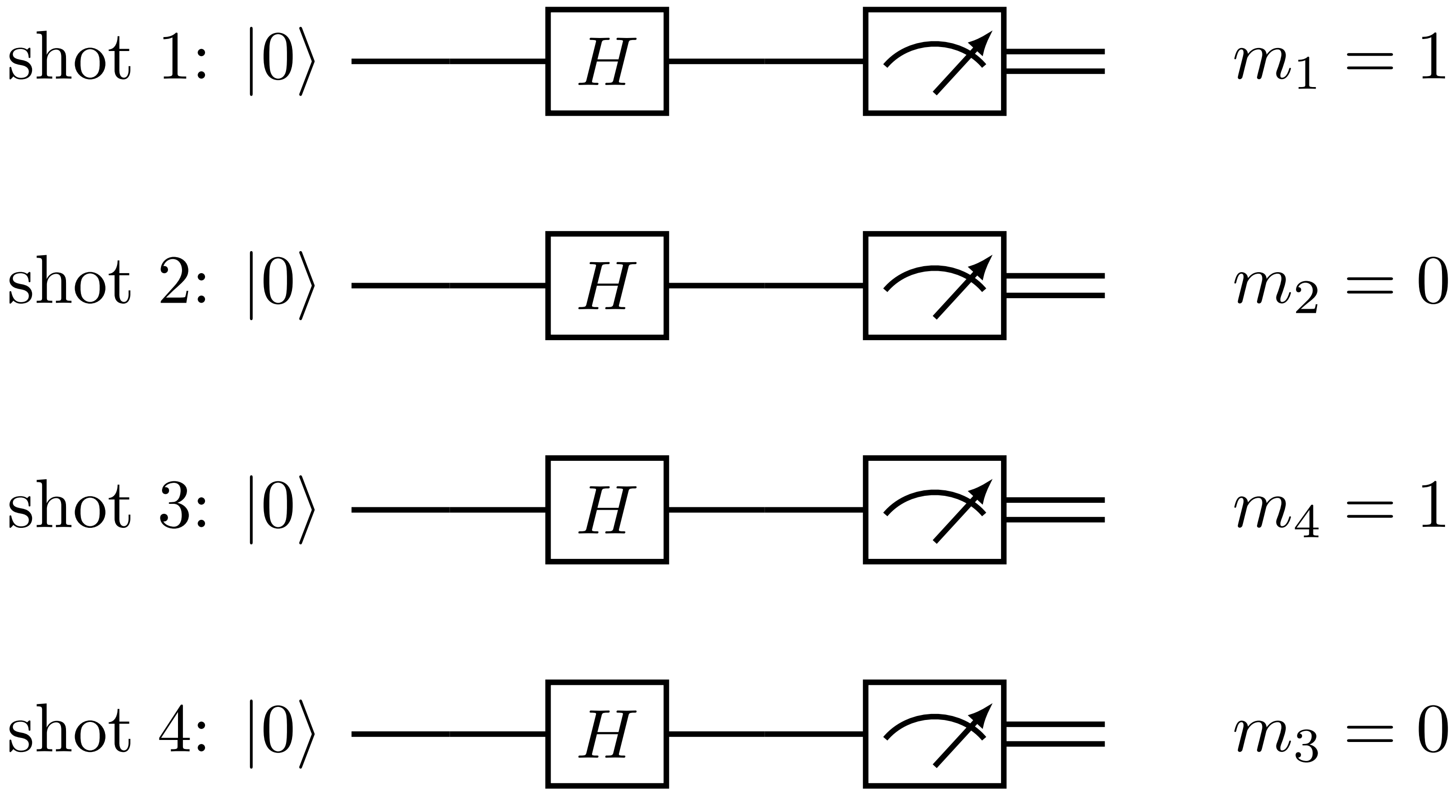
Ideal circuit

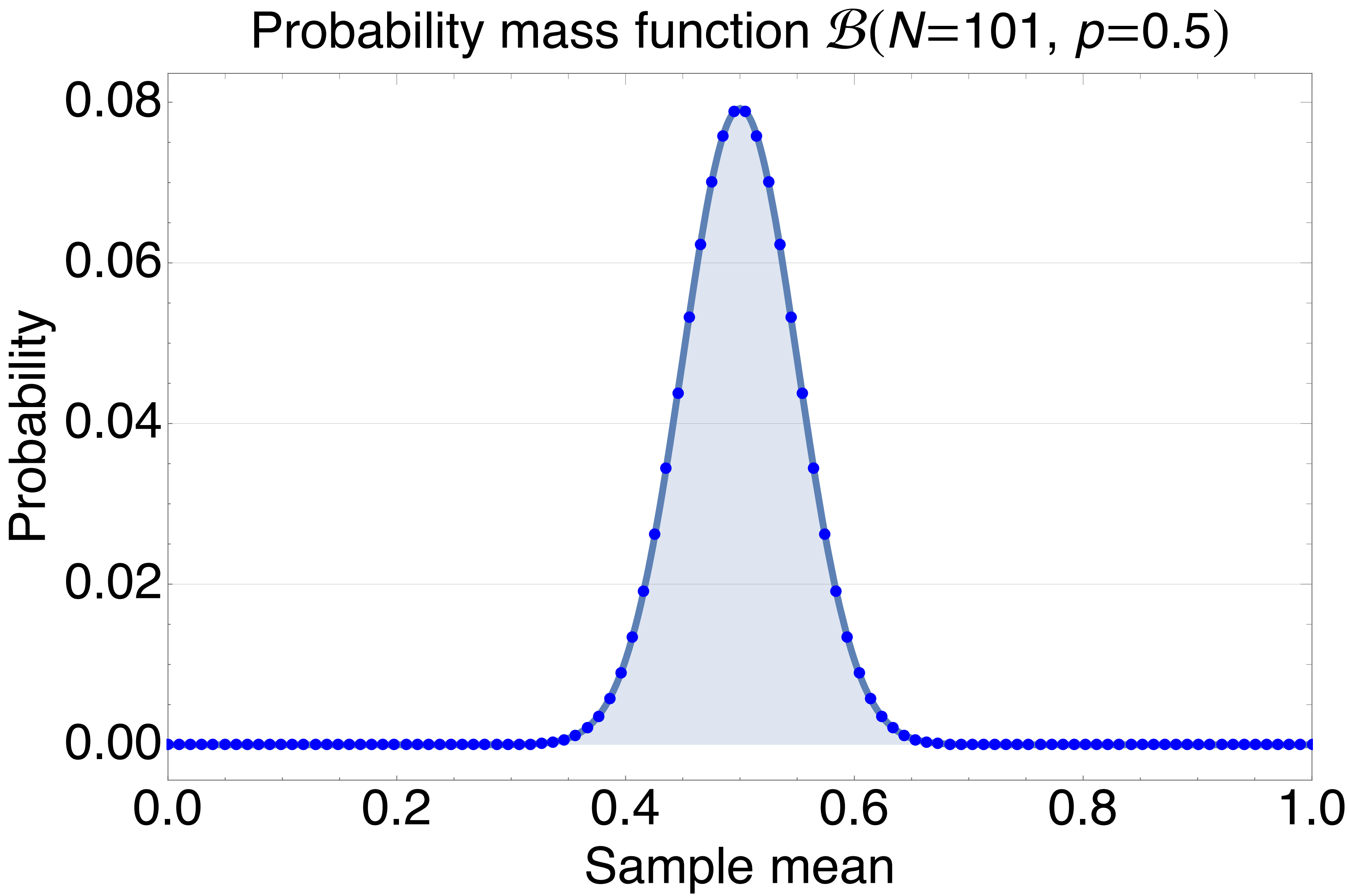




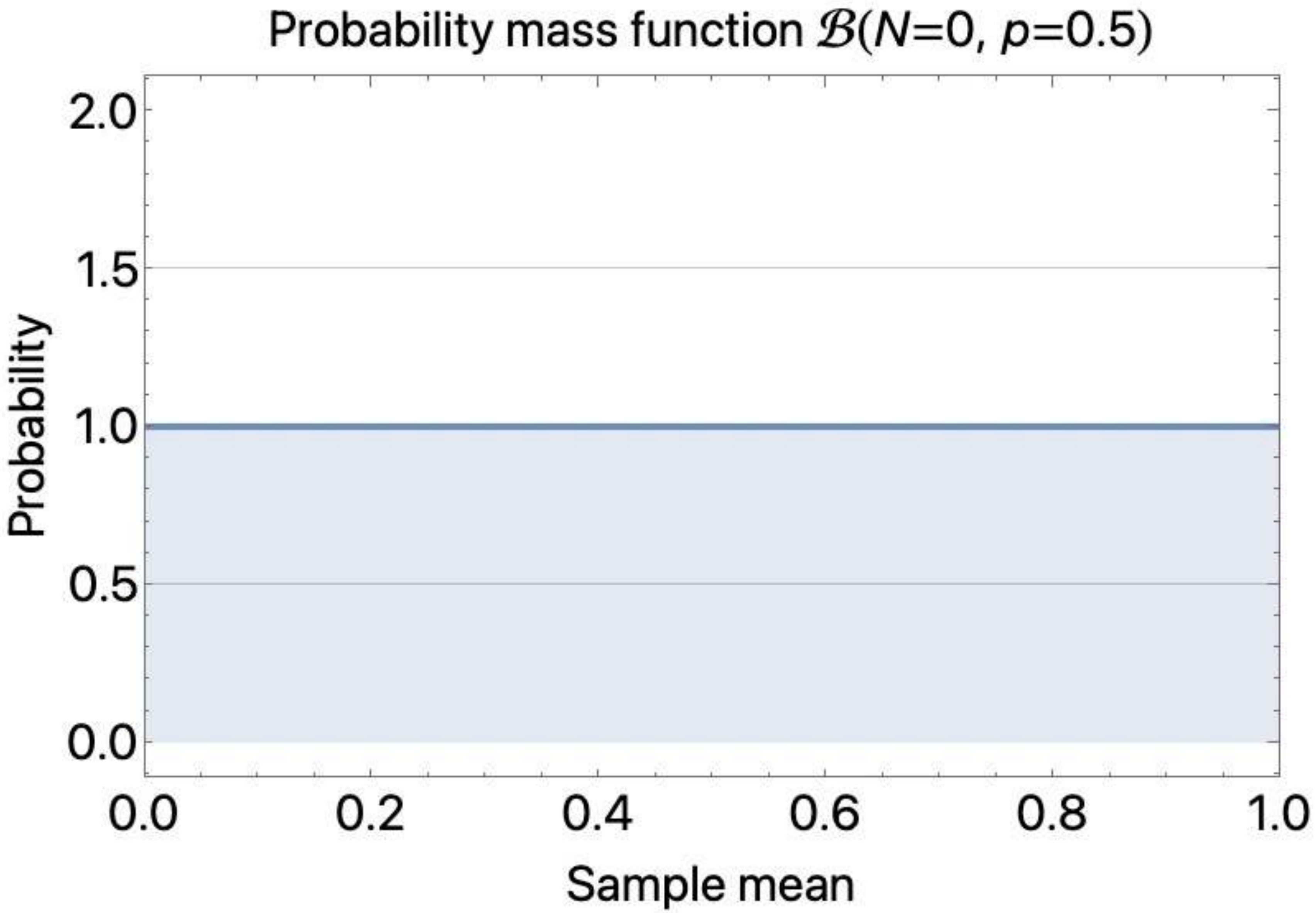
Variance



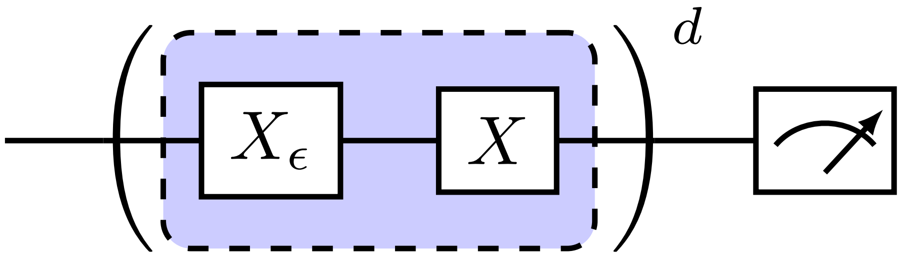

shot 1: $|0\rangle$
H
$m_1 = 1$
shot 2: $|0\rangle$
H
$m_2 = 0$
shot 3: $|0\rangle$
H
$m_4 = 1$
shot 4: $|0\rangle$
H
$m_3 = 0$


Probability mass function $\mathcal{B}(N=101,\ p=0.5)$
0.08
0.06
0.04
0.02
0.00
Probability
0.0
0.2
0.4
0.6
0.8
1.0
Sample mean

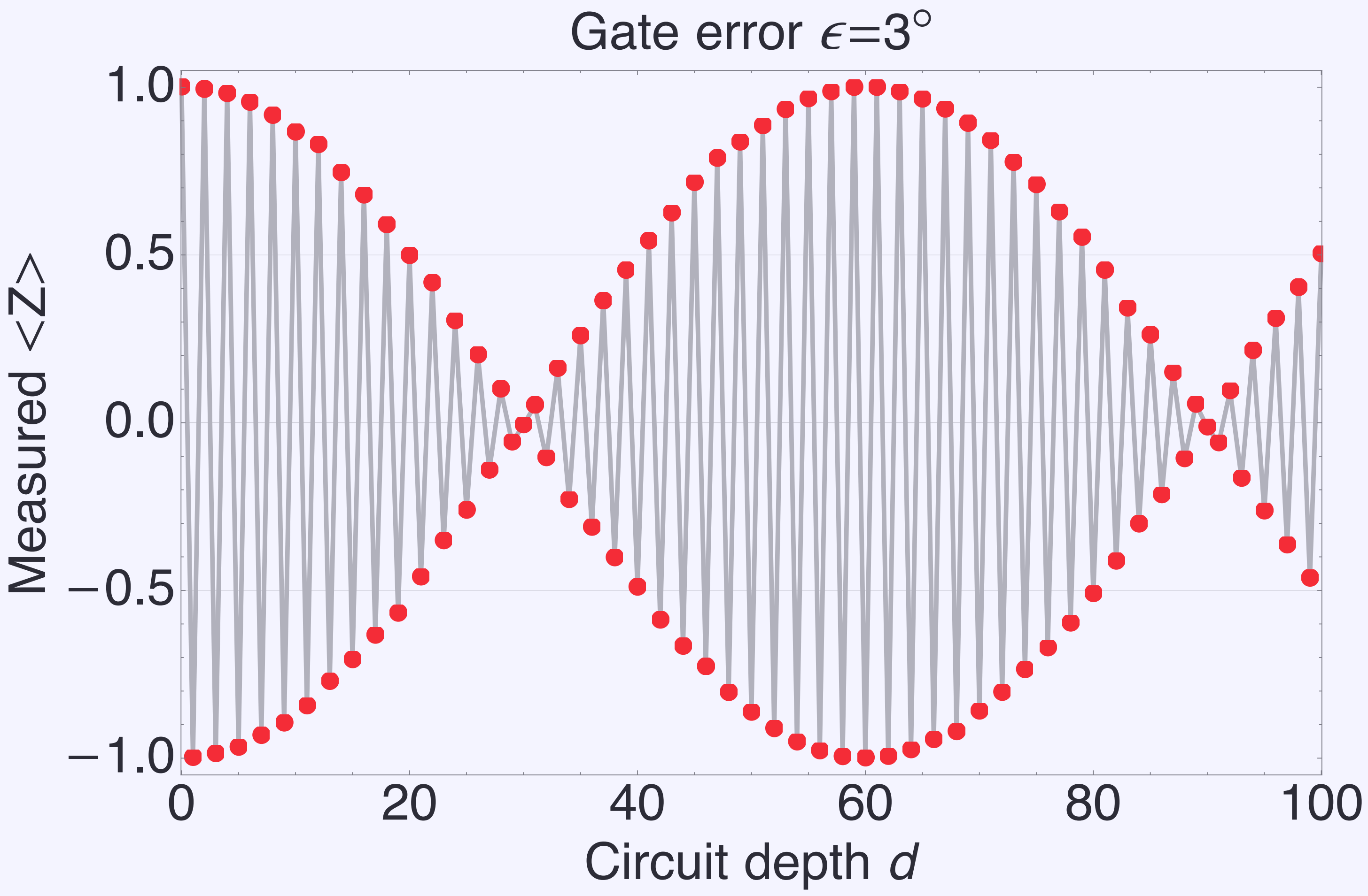
Animation of convergence of shots expectation value and mean



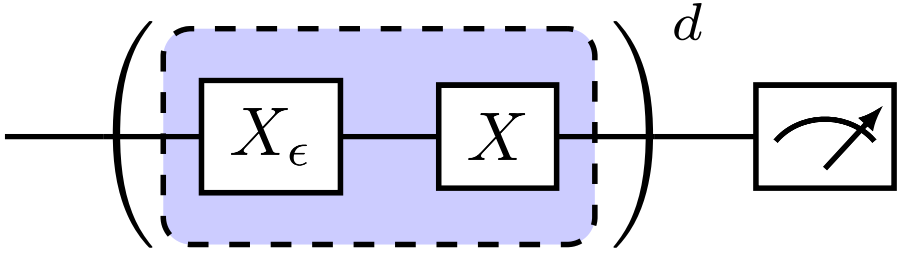
Recall gate error result



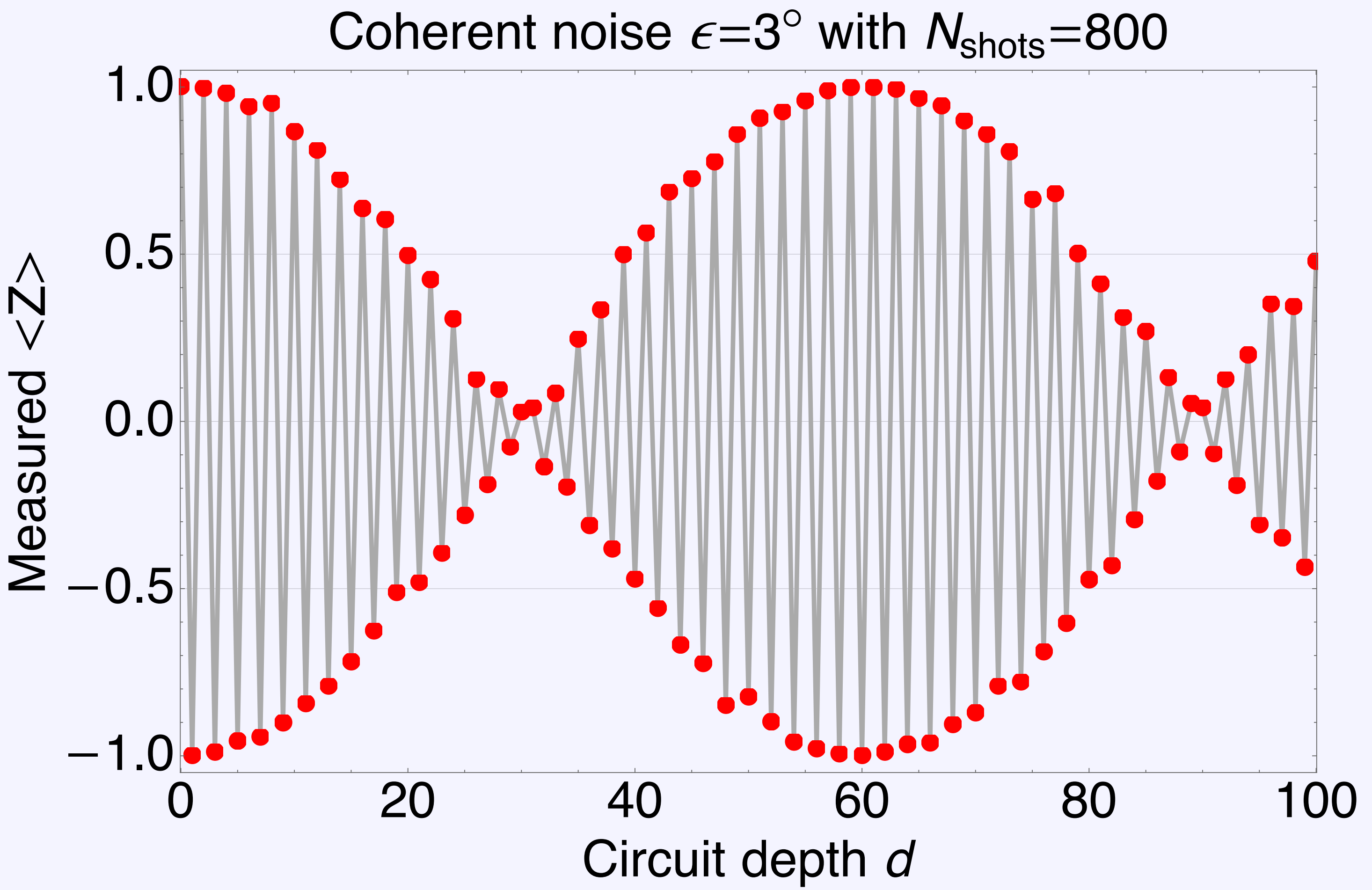
$$\left\langle \tilde{\psi}_f \middle| Z \middle| \tilde{\psi}_f \right\rangle = \cos\left(d\pi + d\epsilon\right)$$



Projection & sampling noise



$$\left\langle \tilde{\psi}_f \middle| Z \middle| \tilde{\psi}_f \right\rangle = \cos\left(d\pi + d\epsilon\right)$$



Dive deeper? Try the following

1. Calculate the following for a qubit
    1. The expectation value of the sample variance for N shots of the observable |1><1|.

       where the sample mean is defined as $$S = \frac{1}{N}\sum_{n=1}^{N} M_n$$

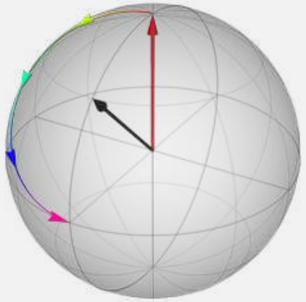
       and the sample variance is defined as $$V = \frac{1}{N}\sum_{n=1}^{N} (M_n - S)^2$$
    2. Is the estimate biased?
    3. The variance of V.
    4. Can you find an expression for an unbiased estimate of the sample variance?

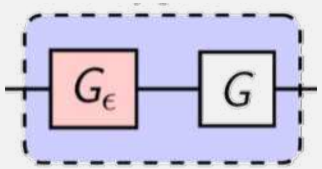
2. What about two qubits?
    1. Can you find the projection operators for the observable ZZ?
    2. Find the probability distribution for the observables ZI, IZ, and ZZ for a general state.

3. If you take 10 shots and find all 10 outcomes to be 1, what is the probability the qubit is in the |0> state? (hint: it’s not zero!)

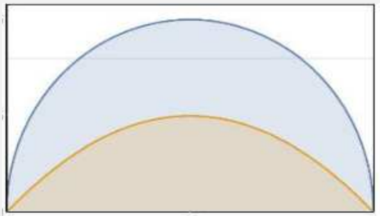
Our road ahead

Coherent

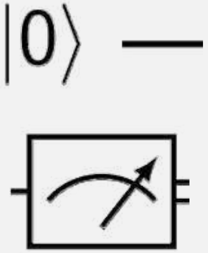


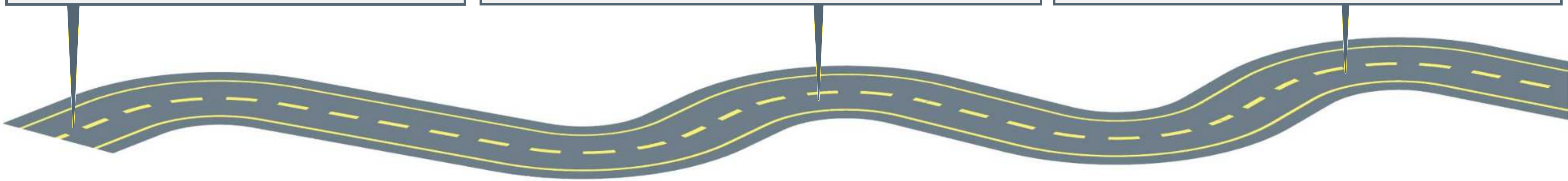


Projection & measurement theory



State preparation & measurement

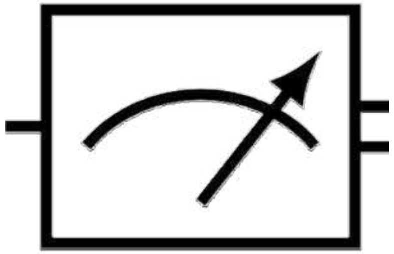


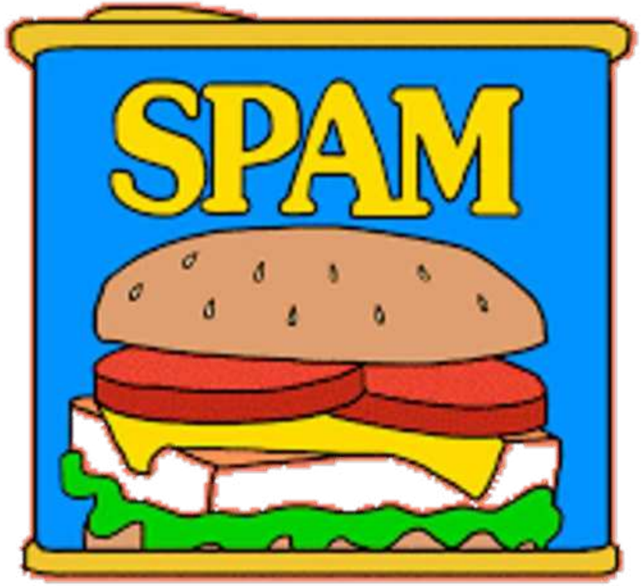


coin toss: flaticon; spam: make it move;
road based on: freepik

State preparation & measurement errors

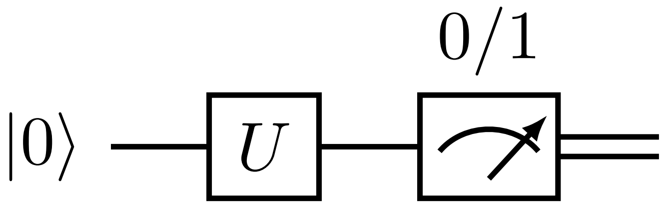
$|0\rangle$ —





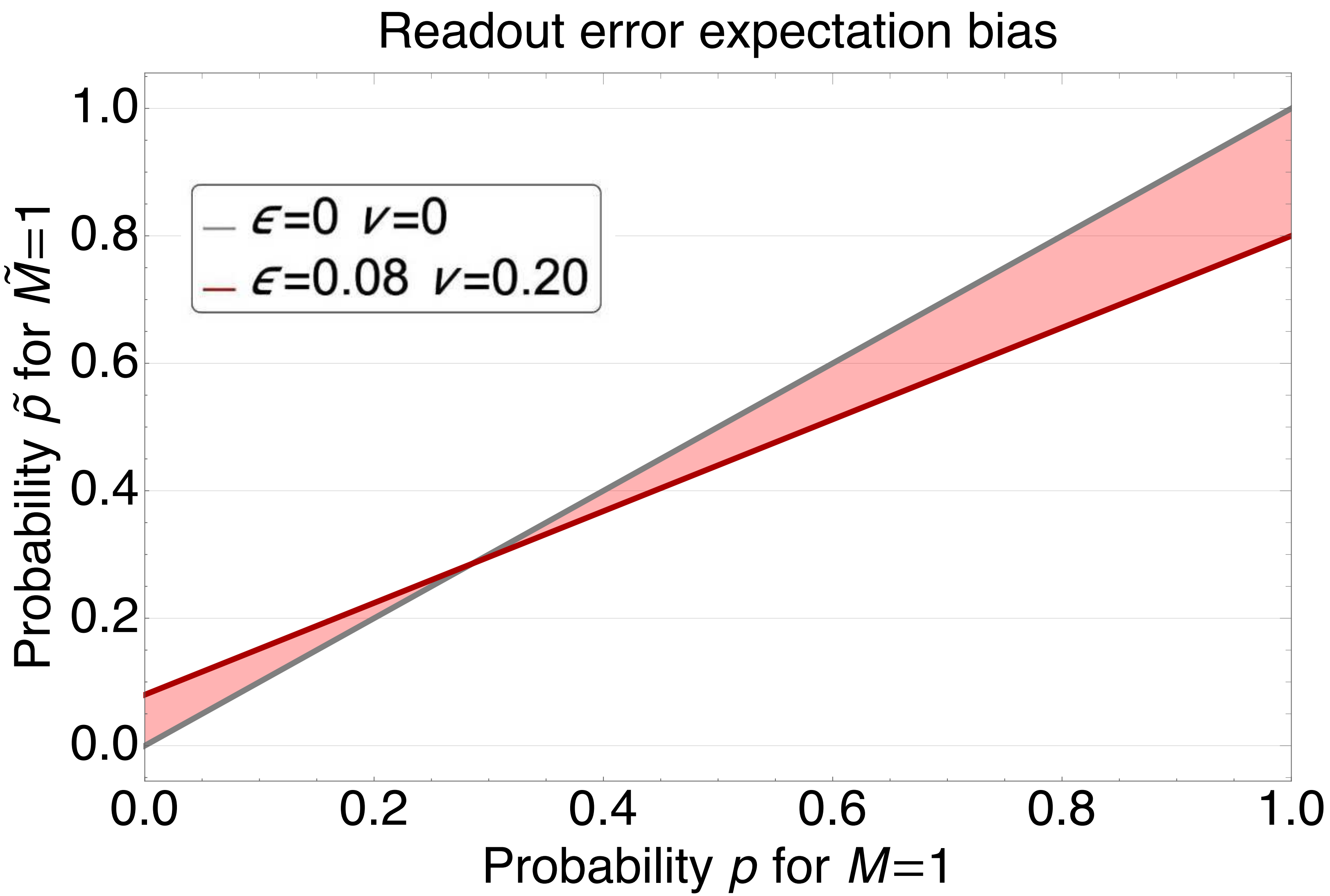
Spam cartoon by Make it move

Qubit example

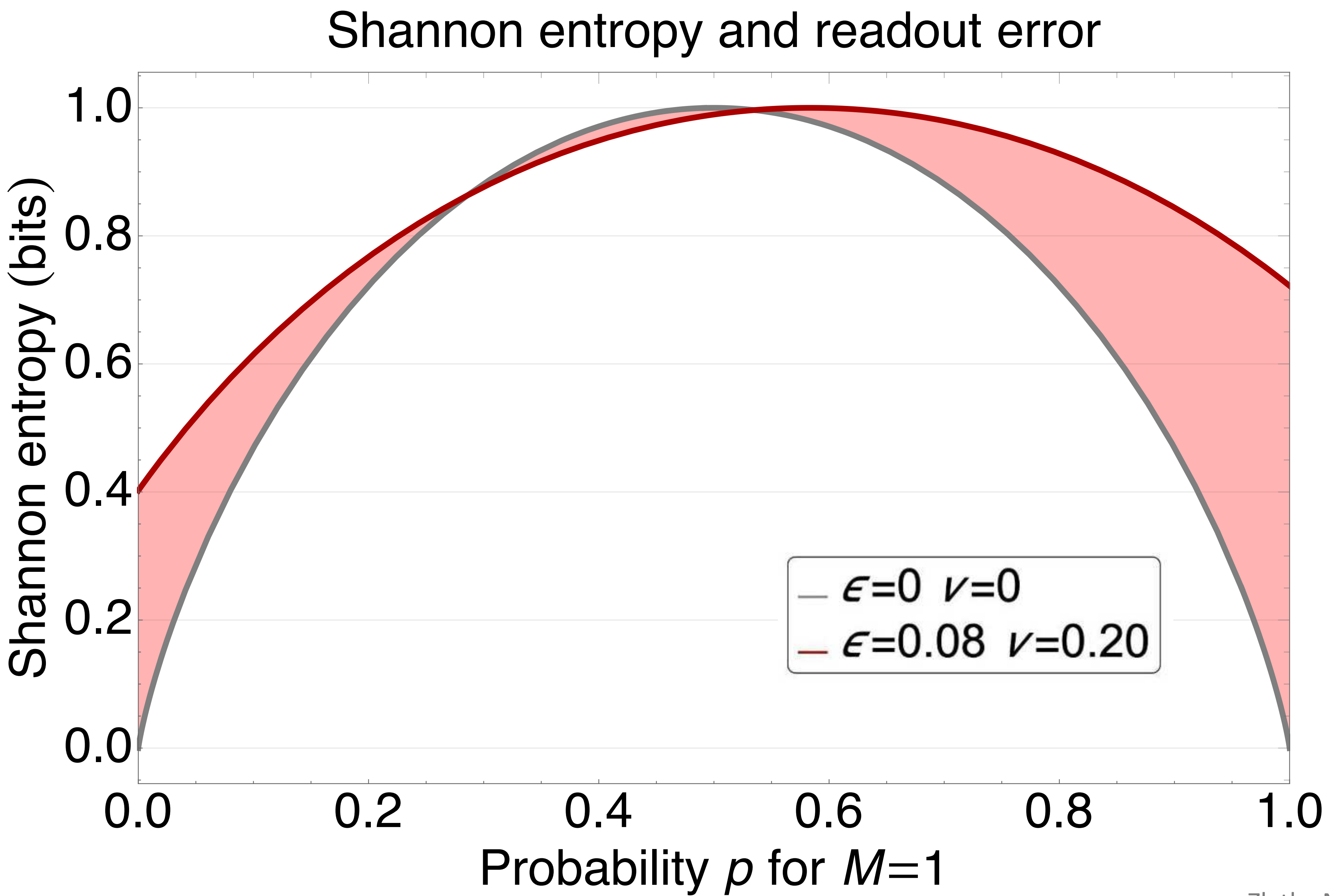


$$\begin{cases} \text{outcome} & \text{probability} & \text{quantum} \\ \mathbb{P}(M=0) = & 1-p & = |\langle 0|\psi\rangle|^2 \\ \mathbb{P}(M=1) = & p & = |\langle 1|\psi\rangle|^2 \end{cases}$$

Readout error

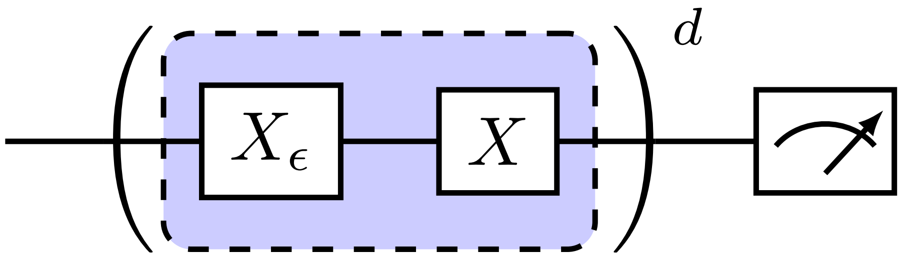


Entropy

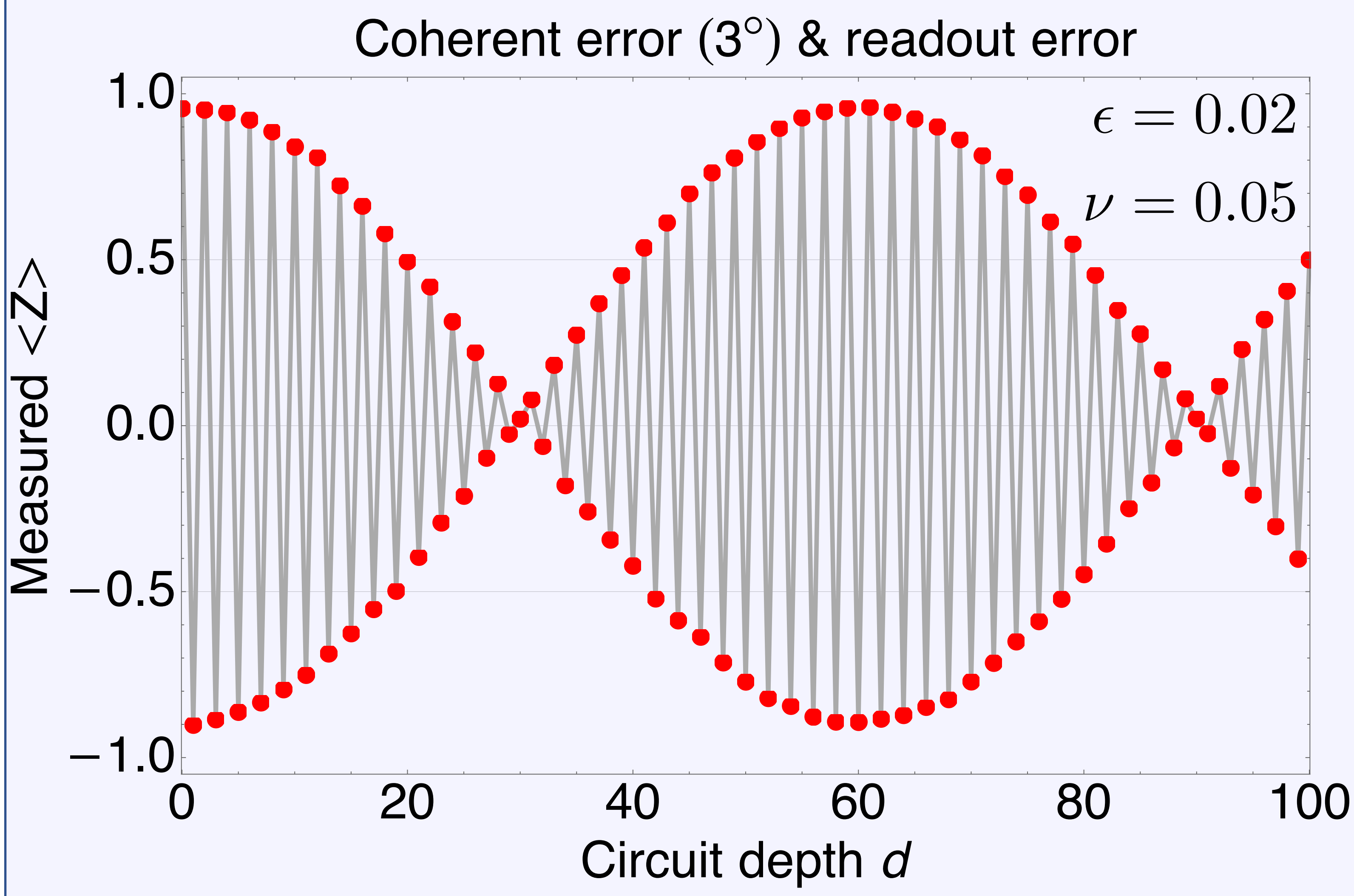


Shannon entropy and readout error
Shannon entropy (bits)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Probability p for M=1
ϵ=0 ν=0
ϵ=0.08 ν=0.20

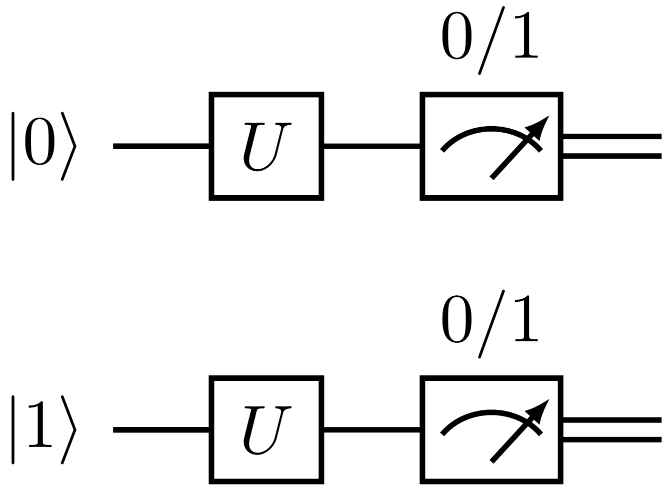
Projection & sampling noise

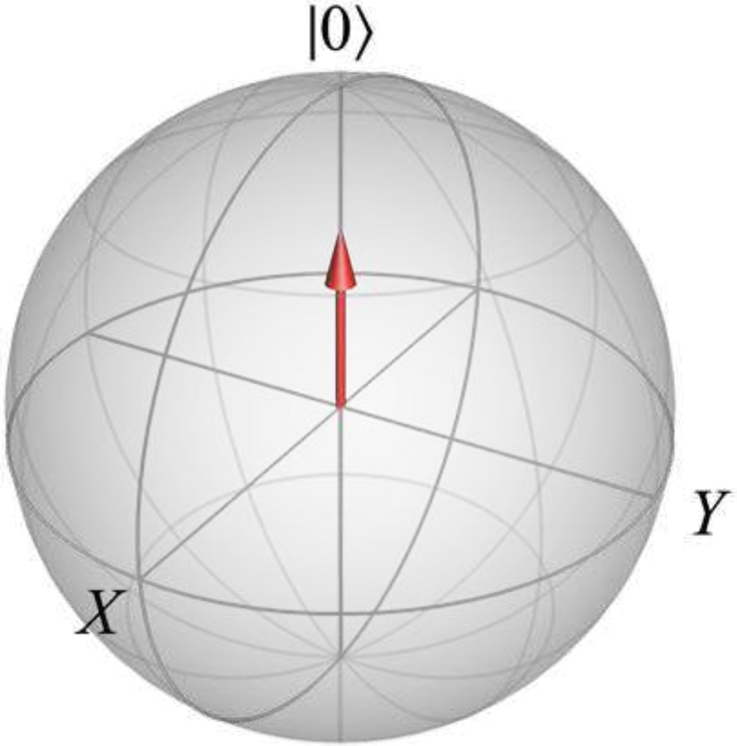


$$A = \begin{array}{c c} & \begin{array}{cc} M=0 & M=1 \end{array} \\ \begin{array}{c} \tilde{M}=0 \\ \tilde{M}=1 \end{array} & \begin{pmatrix} 1-\epsilon & \nu \\ \epsilon & 1-\nu \end{pmatrix} \end{array}$$



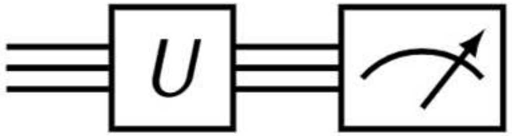
State prep



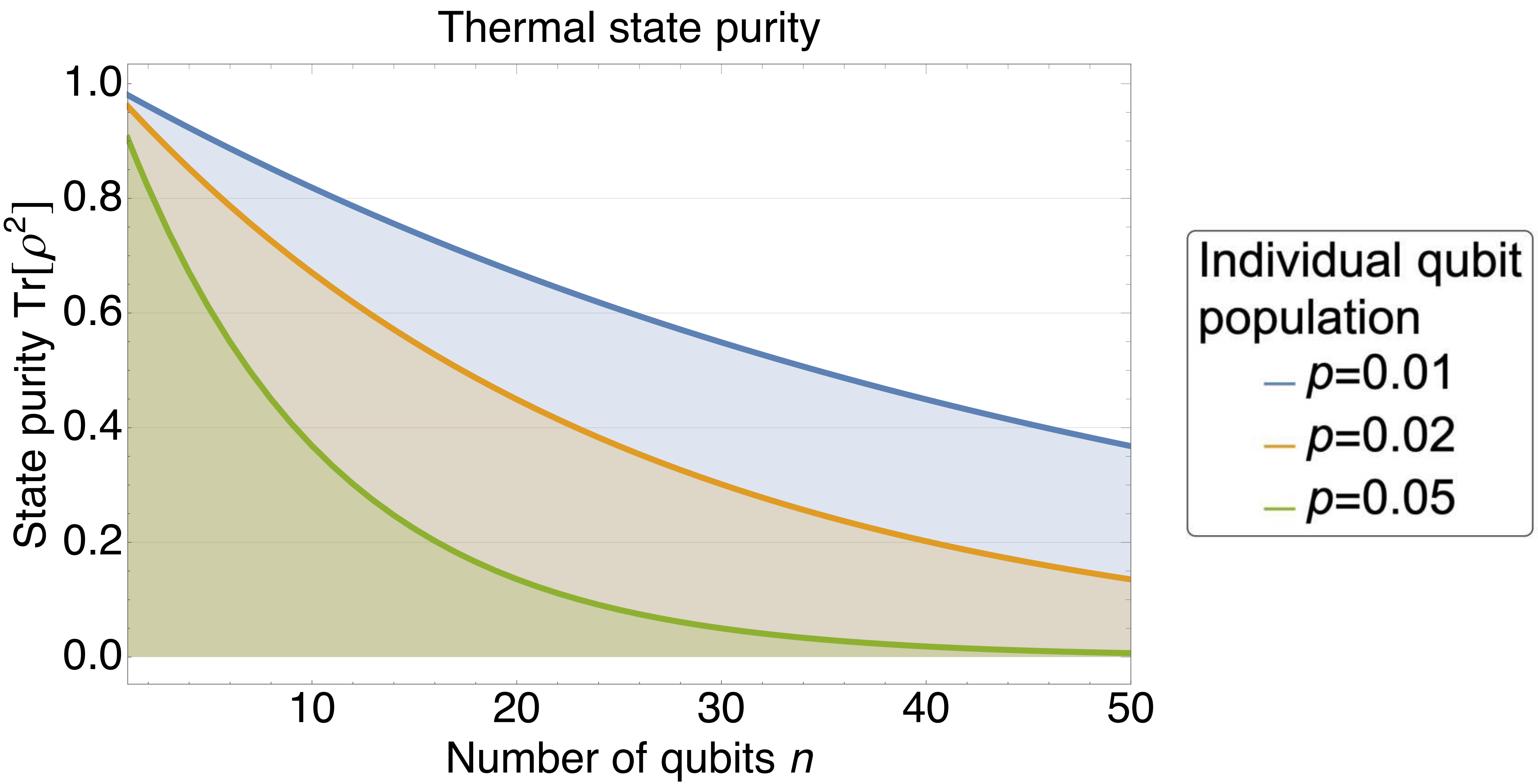


Multiple qubits

$$[(1-p)\,|0\rangle\langle 0| + p\,|1\rangle\langle 1|]^{\otimes n}$$

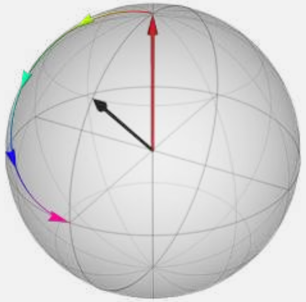


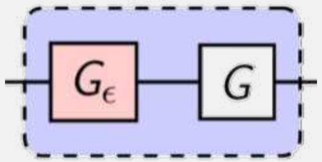
Thermal state



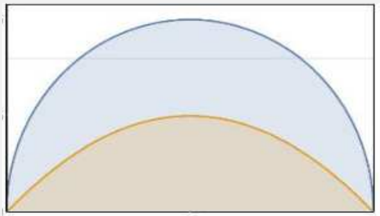
Our road ahead

Coherent

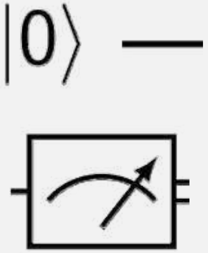


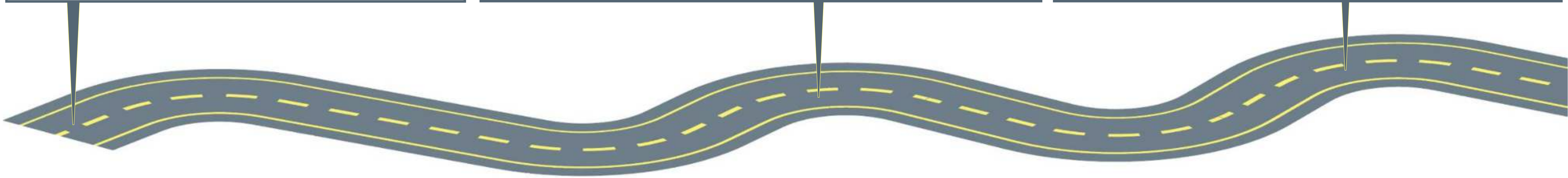


Projection & measurement theory

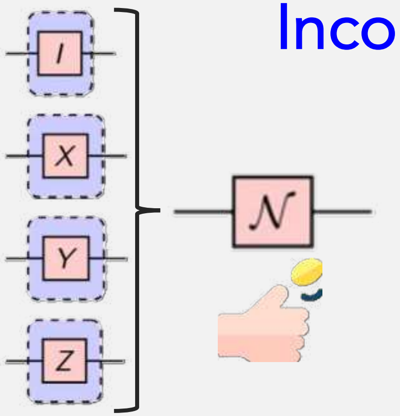


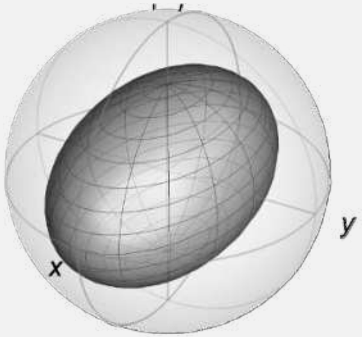
State preparation & measurement



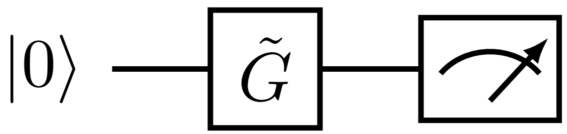


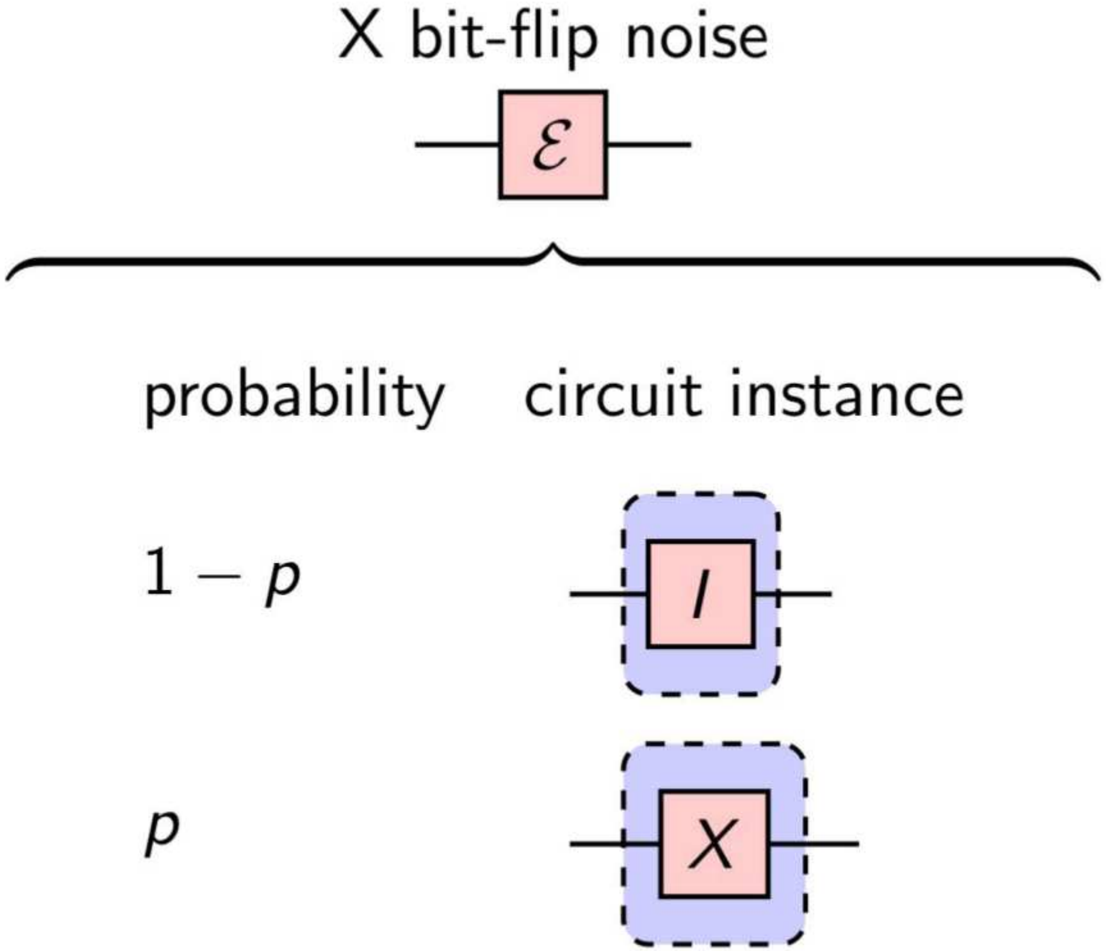
Incoherent

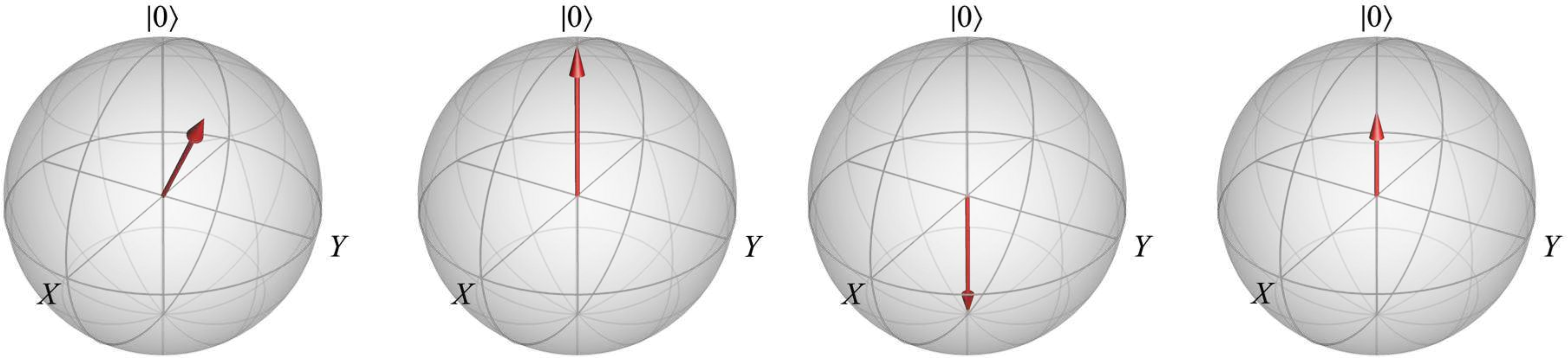




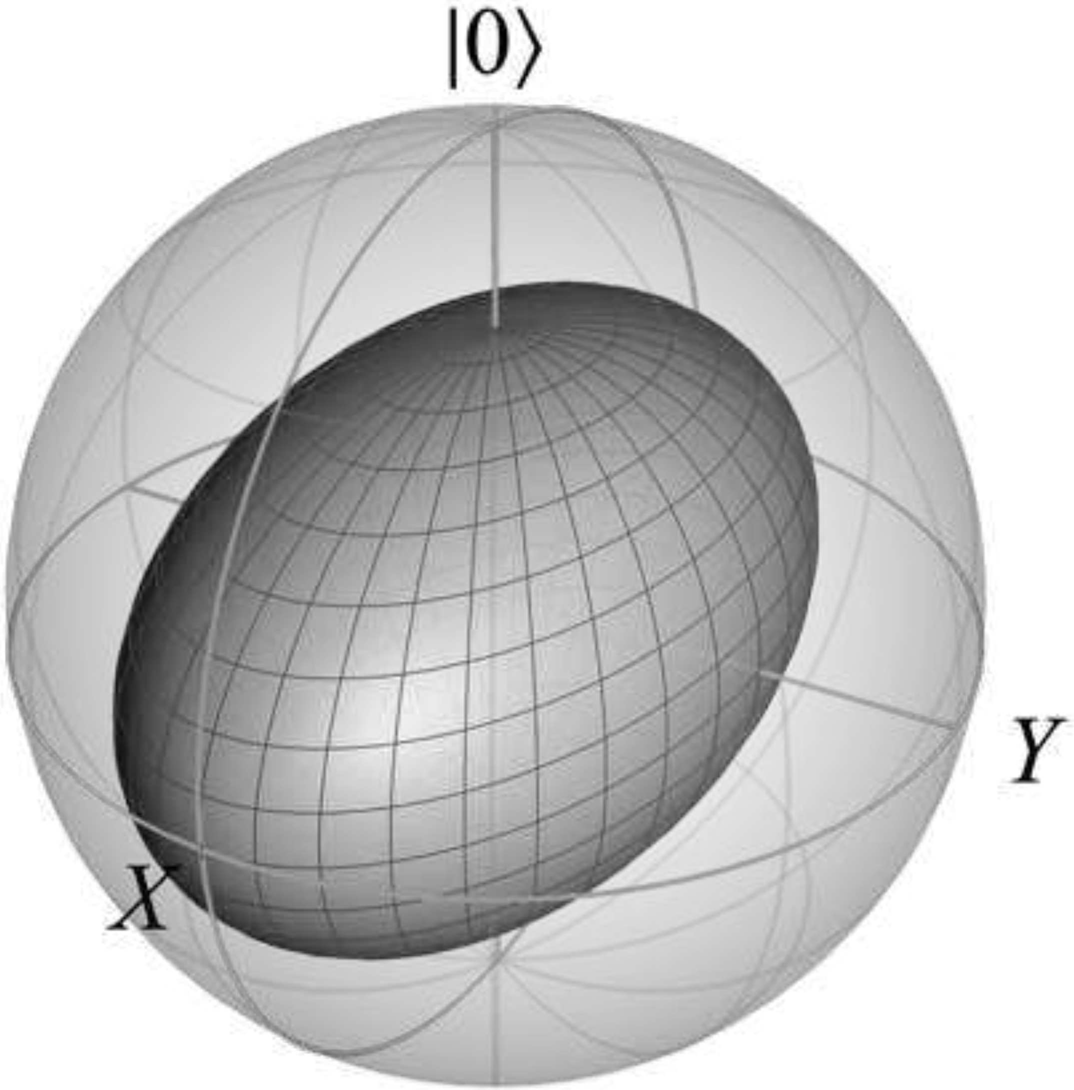
coin toss: flaticon; spam: make it move;
road based on: freepik

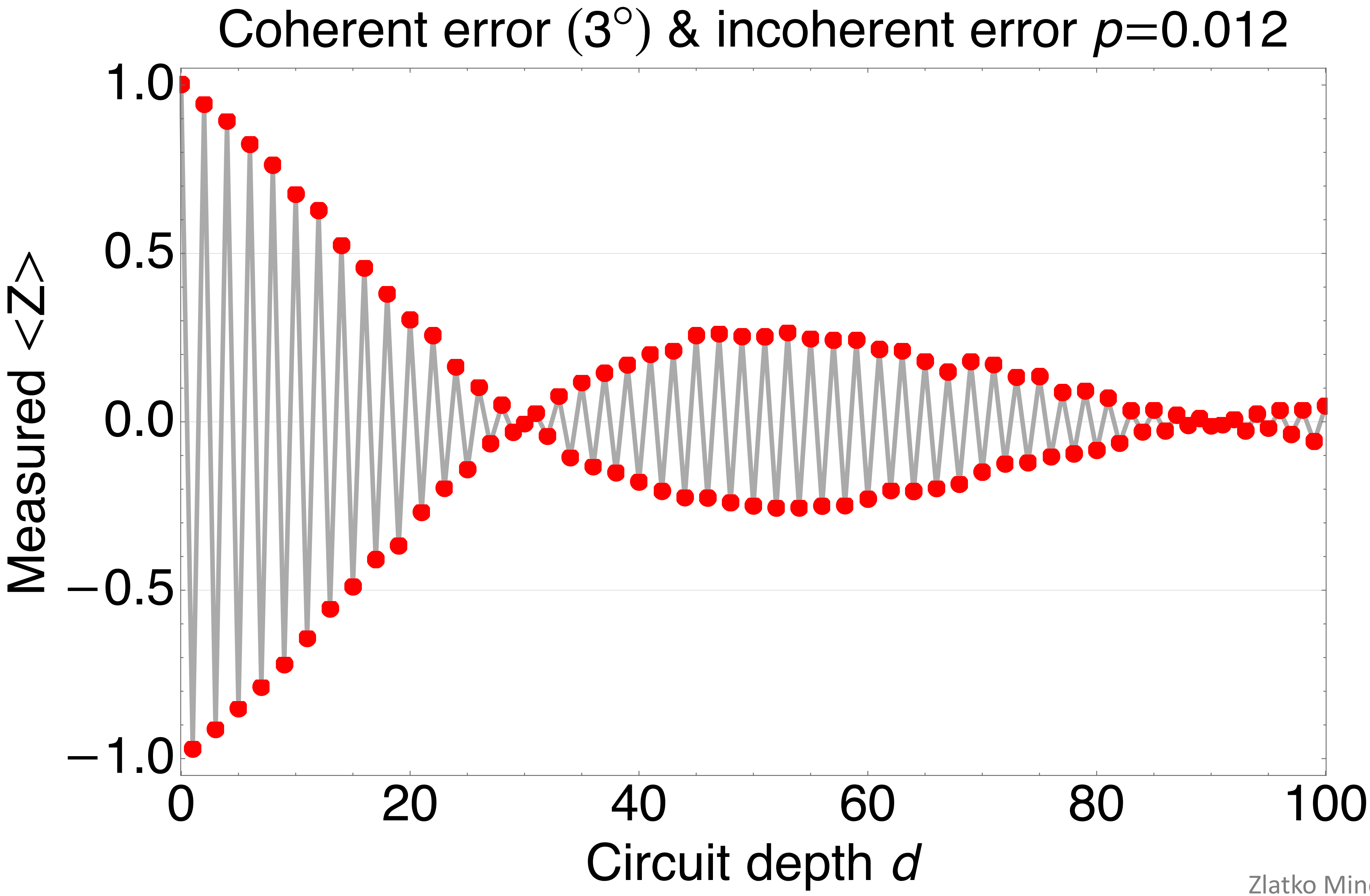

$|0\rangle$
$\tilde{G}$


X bit-flip noise
$\mathcal{E}$
probability
circuit instance
$1 - p$
I
p
X

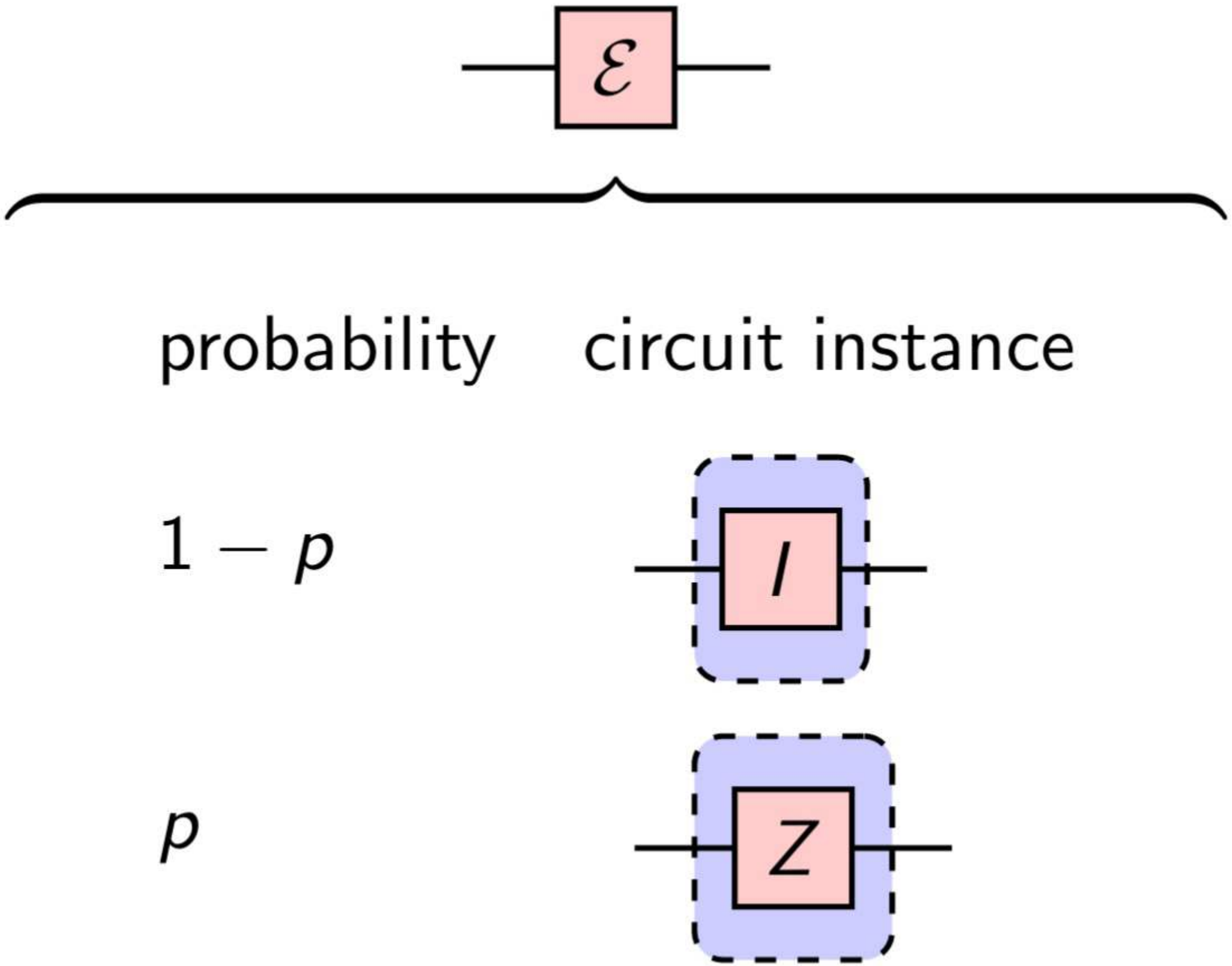

|0⟩
X
Y
|0⟩
X
Y
|0⟩
X
Y
|0⟩
X
Y

Bit-flip noise

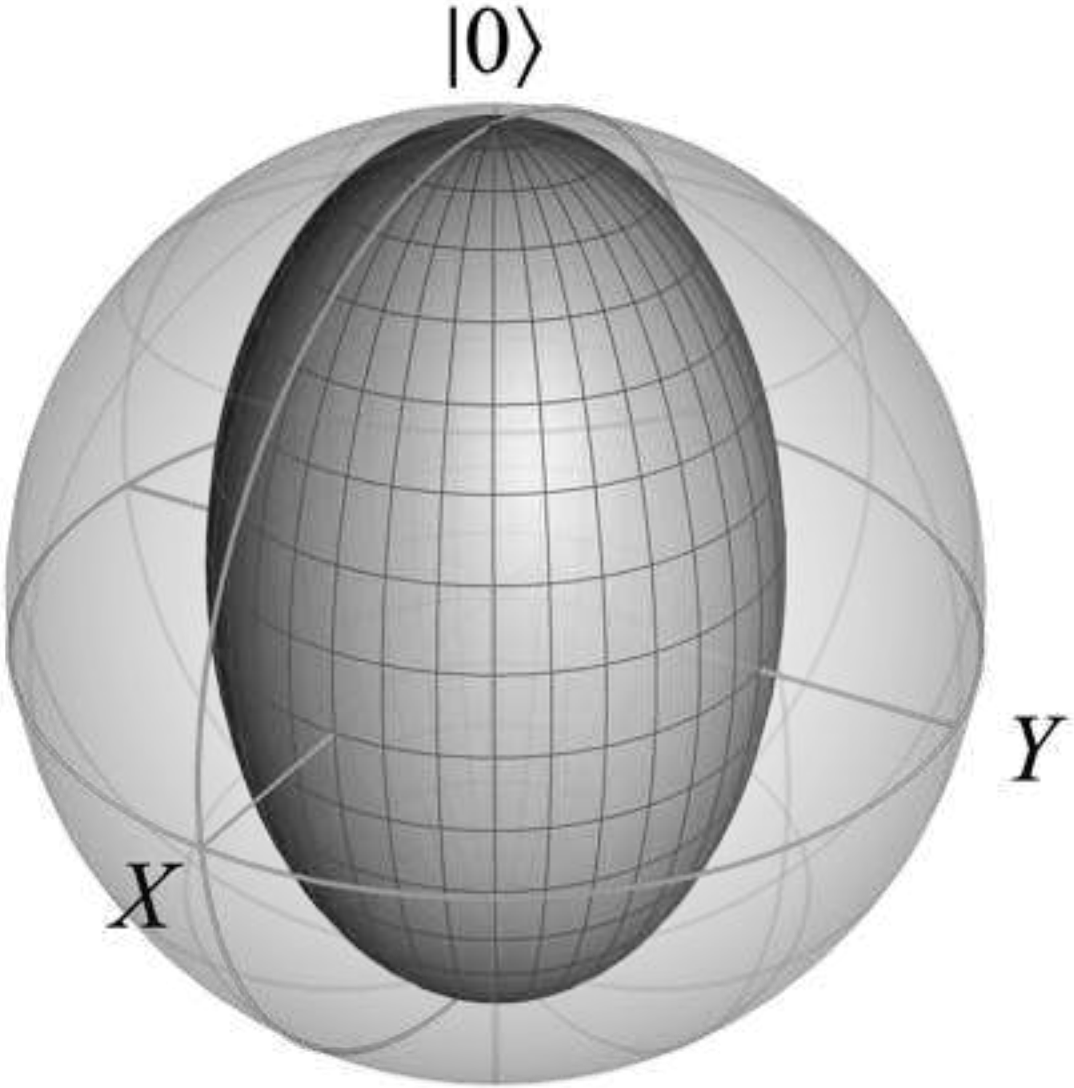



Coherent error (3°) & incoherent error p=0.012
Measured <Z>
1.0
0.5
0.0
−0.5
−1.0
0
20
40
60
80
100
Circuit depth d

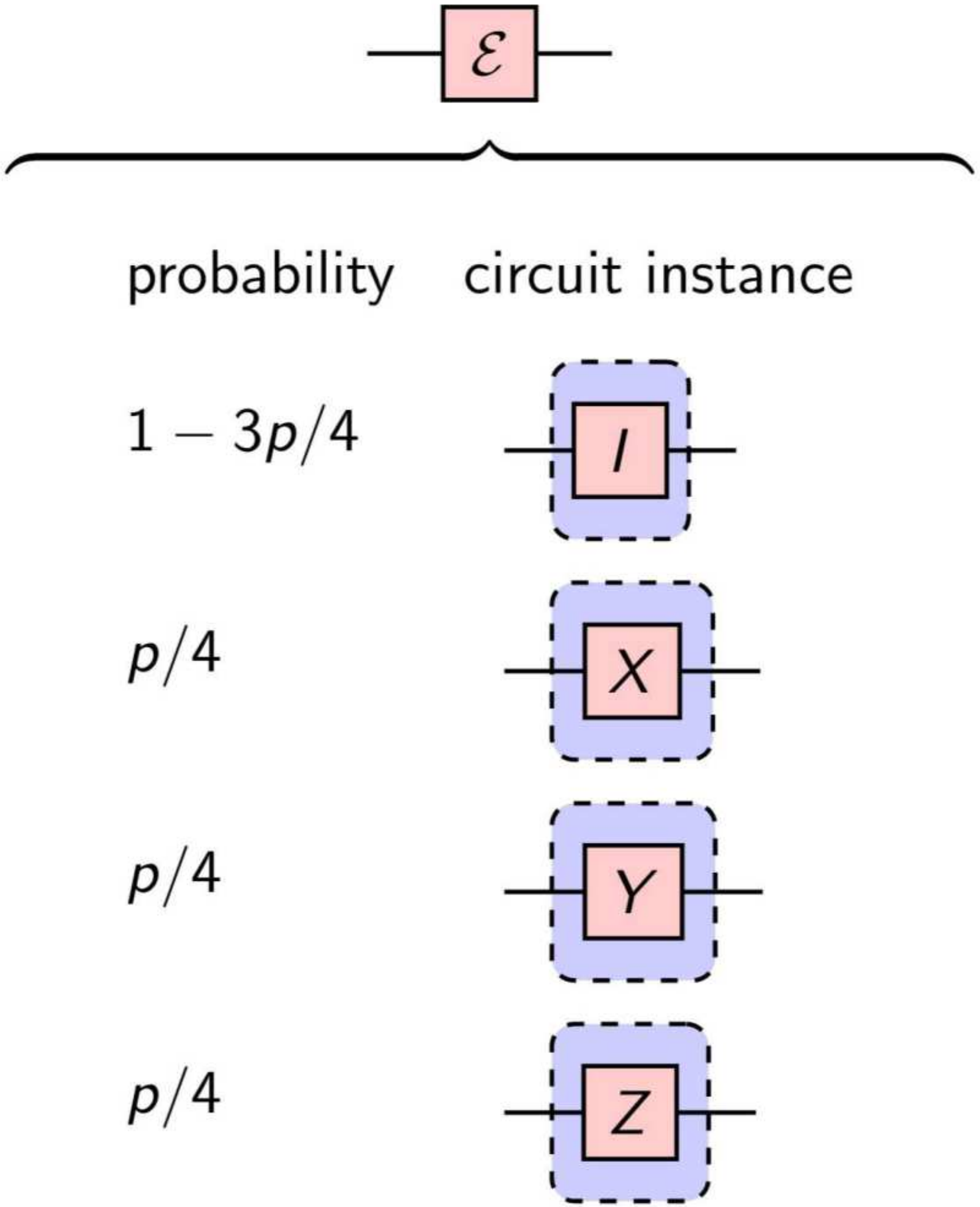
Phase noise



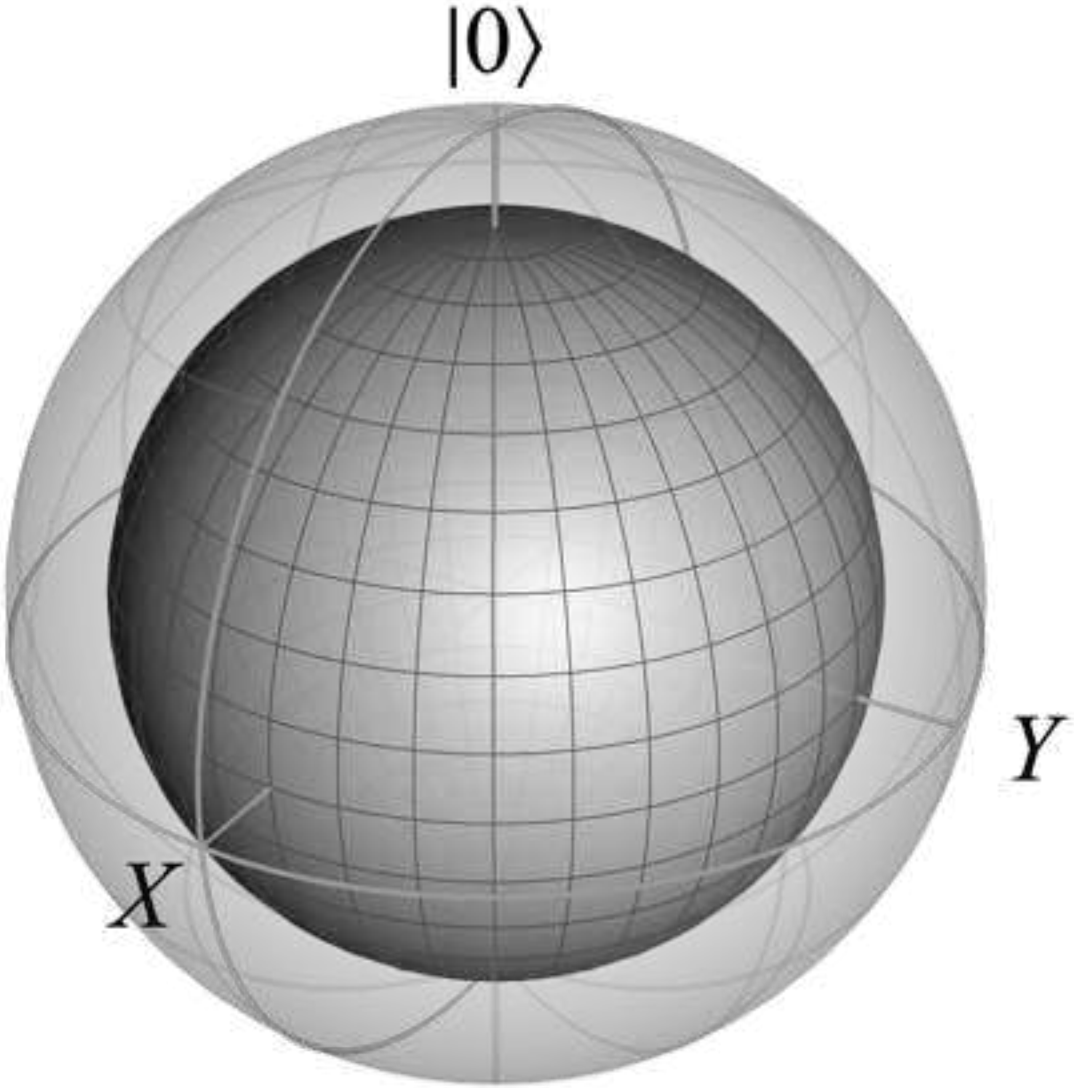
Phase damping noise



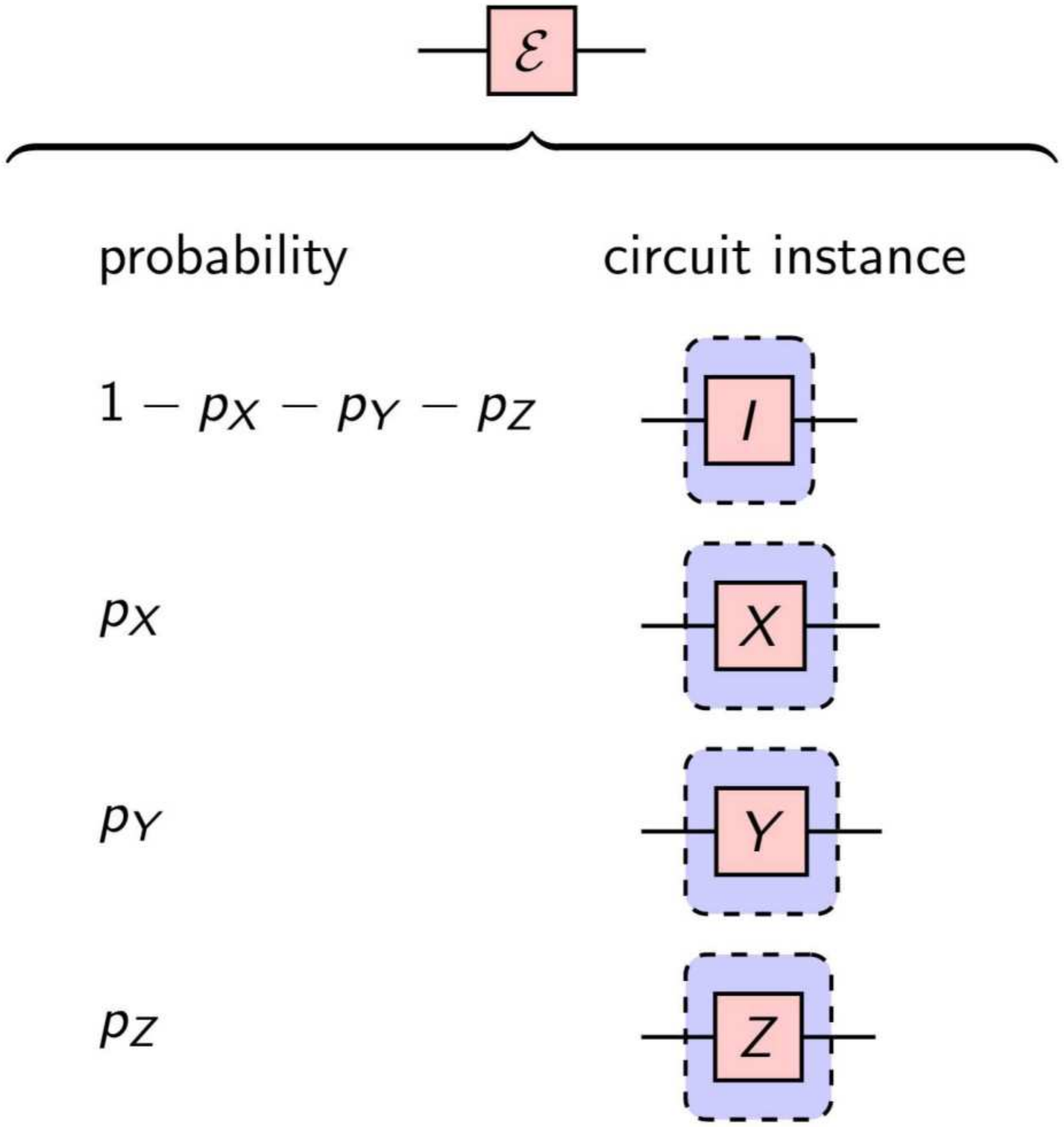
Depolarizing noise



Depolarizing noise

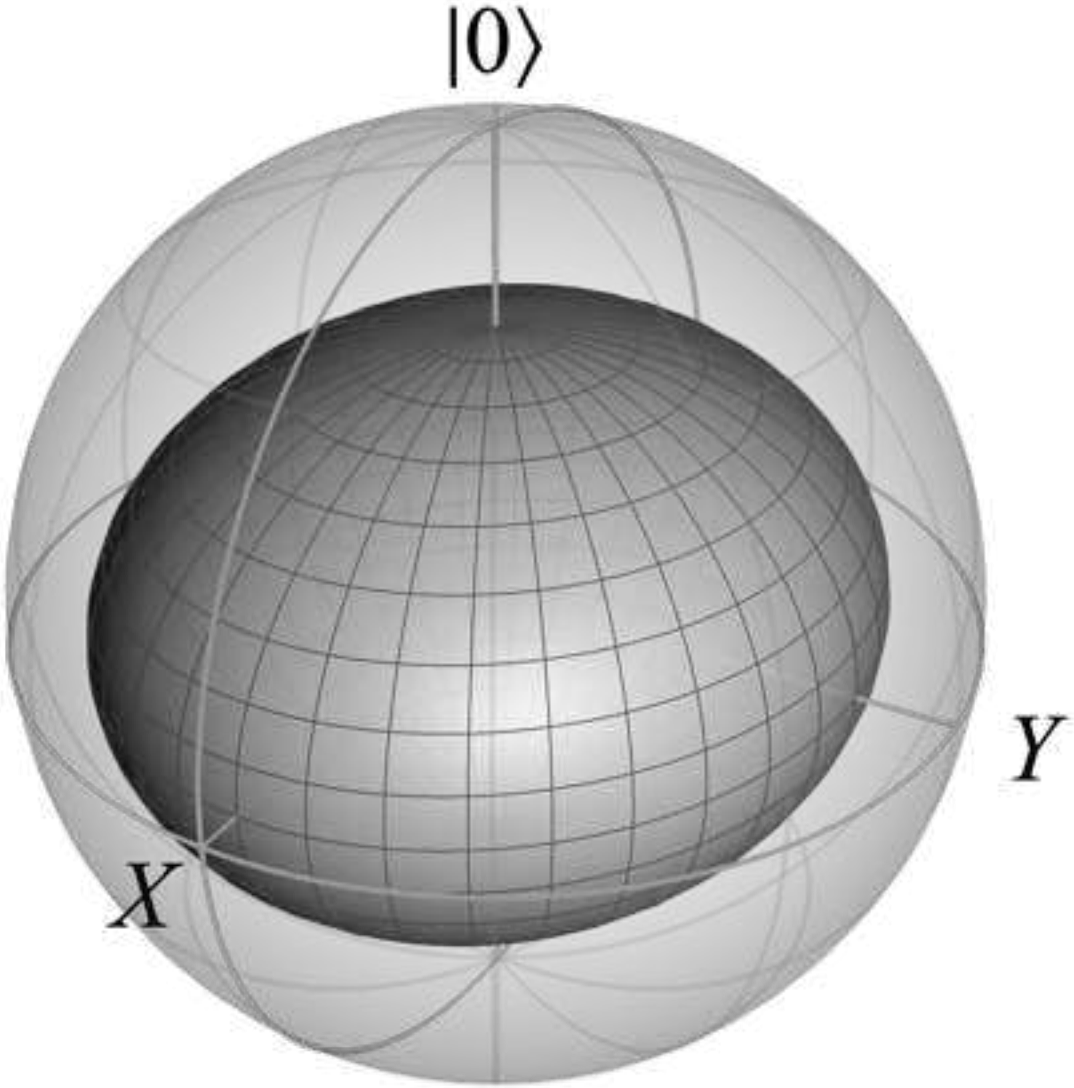


Stochastic Pauli noise

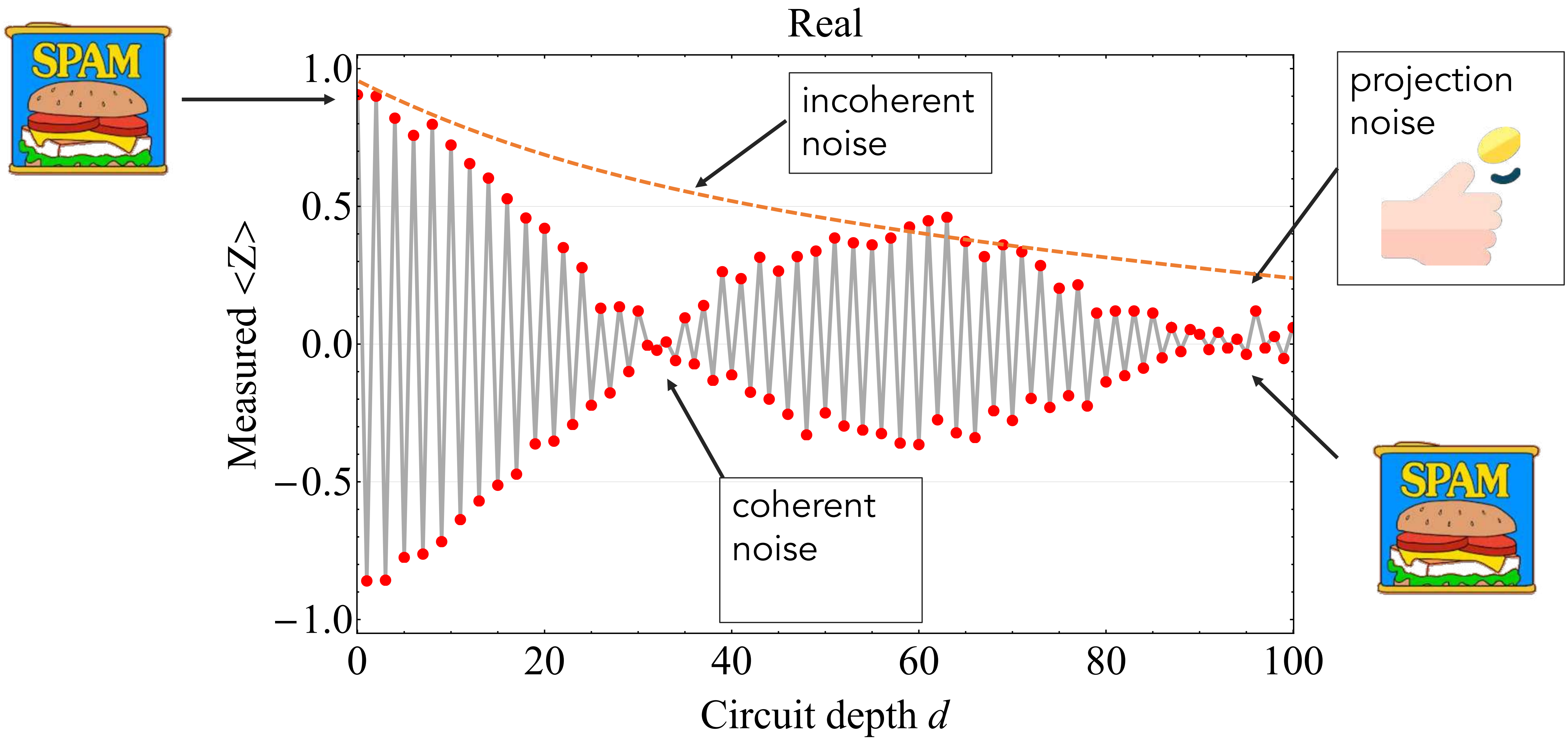


Stochastic Pauli Noise

$p_x = 0.1$
$p_y = 0.2$
$p_z = 0.4$

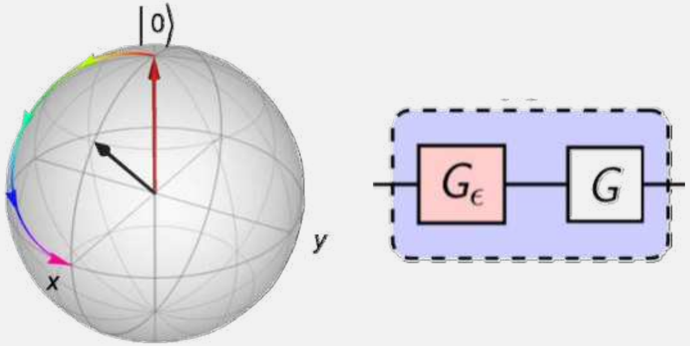


Elements of noise



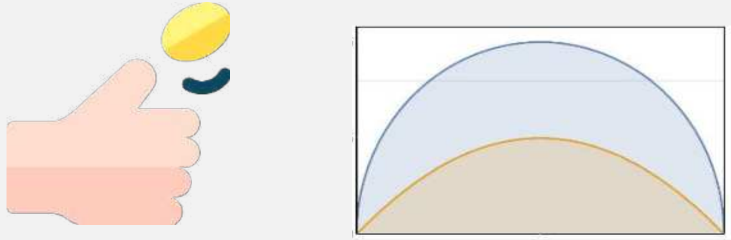
The journey behind us

Coherent noise

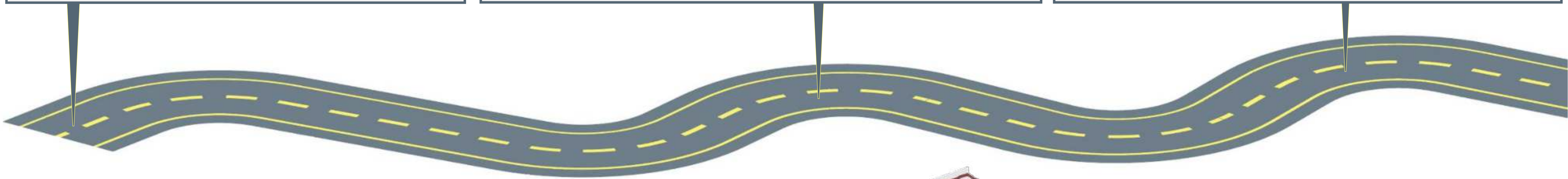


Quantum measurement theory

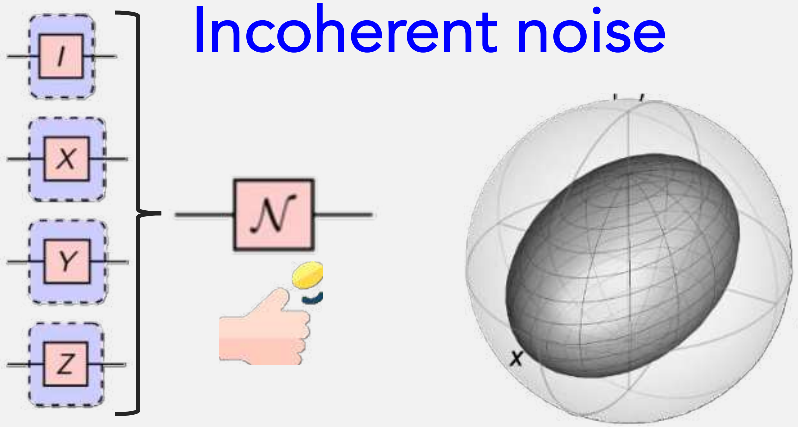
Projection & sampling noise



State prep & measure



Incoherent noise



coin toss: flaticon; spam: make it move;
road based on: freepik

Next steps

Lab work with Qiskit

Run experiments on real devices

Check out references, problems given in the lecture, dangerous bends

Stay in touch

Thank you!

Zlatko K. Minev

The important thing is not to stop questioning. Curiosity has its own reason for existence.

One cannot help but be in awe when he contemplates the mysteries of eternity, of life, of the marvelous structure of reality.

It is enough if one tries merely to comprehend a little of this mystery each day.

Albert Einstein

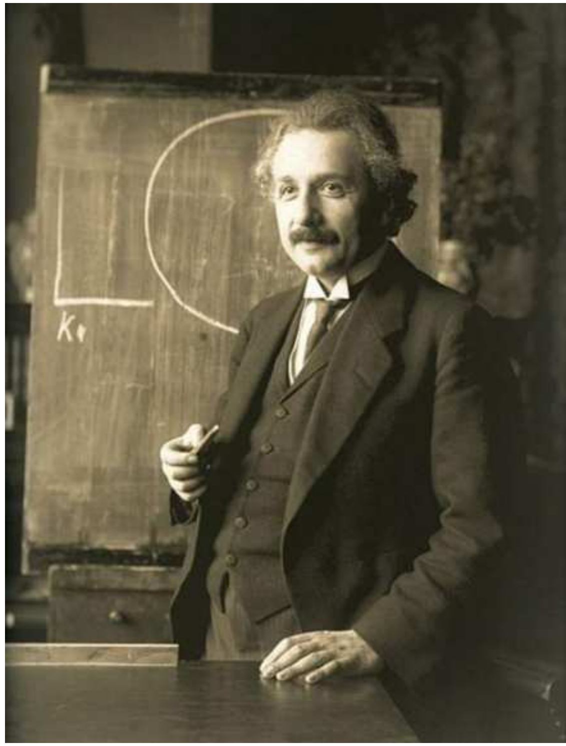


Photo: F. Schmutzer